

COMPUTATIONAL FEASIBILITY STUDY OF CAPSULE-BASED SELF-HEALING EPOXY COMPOSITES MANUFACTURED BY DIRECT INK WRITING (DIW) 3D PRINTING

An Project Report

Submitted by

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2026

BONAFIDE CERTIFICATE

This is to certify that the project report titled “Computational Feasibility Study of Capsule-Based Self-Healing Epoxy Composites Manufactured by Direct Ink Writing (DIW) 3D Printing” is a Bonafide record of the internship work carried out by Aadil Ahmad Magray, Krupank Gowda, Darshan R Department of Aeronautical Engineering, Acharya Institute of Technology, during the internship at the Department of Mechanical Engineering, National Institute of Technology, Tiruchirappalli, under my supervision.

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ABSTRACT

Capsule-based self-healing composites — in which microencapsulated healing agent is embedded within a structural polymer matrix and released autonomously upon crack formation — have been studied extensively since their introduction by White et al. (2001), with demonstrated improvements in fracture toughness, fatigue life, and damage recovery across numerous composite systems. Separately, additive manufacturing, and Direct Ink Writing (DIW) in particular, has matured into a viable route for producing structurally capable polymer and fibre-reinforced composites. However, a recent review by Bekas et al. (2019) explicitly identifies that the integration of multiple, sophisticated functionalities — including self-healing — into 3D-printed composites remains comparatively underdeveloped, and this gap is independently reinforced by Jiang et al. (2023) and Sanders et al. (2019), the latter of whom note that DIW specifically requires substantially more rheological optimisation than alternative printing methods such as stereolithography. It has, until now, remained unresolved whether capsule-based self-healing agents can physically survive the mechanical conditions of DIW printing without rupturing prematurely during manufacture itself.

This project addresses that gap through a purely computational feasibility study, structured around the three mechanical prerequisites that must each hold true for capsule-based self-healing to function in a DIW-printed epoxy composite: the capsules must survive the shear stresses of nozzle extrusion during printing; they must rupture reliably when a crack approaches during service; and the released healing agent must flow into and fill the resulting crack. Three finite-element simulations were developed to evaluate each prerequisite in turn. Simulation 1, built in COMSOL Multiphysics as a 2D axisymmetric laminar flow model of a DIW nozzle, evaluated the shear stress experienced by an embedded urea-formaldehyde (UF) microcapsule across a range of inlet pressures (0.5–3.0 bar) and capsule radii (0.10–0.25 mm). Simulation 2, built in ANSYS Mechanical as a three-dimensional epoxy block containing discrete spherical capsule inclusions, evaluated the stress concentration factor (SCF) around capsules under tensile service loading across capsule size and concentration parametric studies. Simulation 3, implemented in COMSOL as a one-dimensional time-dependent ordinary differential equation based on the Washburn capillary flow equation, evaluated the penetration of dicyclopentadiene (DCPD) healing agent into cracks of varying width (5–50 μm).

Results show that maximum shear stress on the capsule surface during printing reached only 0.080 MPa at the highest pressure tested, a safety factor of approximately 500 times below

the literature-reported UF shell failure strength of 40 MPa, indicating that capsule survival during DIW printing is mechanically assured across the full range of conditions studied. The stress concentration analysis yielded an SCF of approximately 1.57 at the standard three-capsule configuration, consistent with the expected behaviour of spherical fluid-filled inclusions reported in the literature, with capsule concentration — rather than capsule size — identified as the dominant variable governing stress concentration; concentrations beyond three capsules per unit volume produced a sharp rise in stress due to capsule–capsule interaction. The capillary flow analysis showed that healing agent penetration follows the expected square-root-of-time relationship and is capable of filling a 1 mm crack in under 0.036 seconds across all crack widths tested. All three simulations were independently cross-validated using Random Forest regression and polynomial regression machine learning models trained on the parametric simulation data, achieving R^2 scores between 0.93 and 0.9996, with feature-importance analysis confirming — independently of the FEM results — that capsule concentration accounts for approximately 98% of the variation in stress concentration, against approximately 2% for capsule size. Mesh convergence, dimensionless number analysis (Reynolds, Capillary, Stokes, Bond, and Ohnesorge numbers), and analytical validation against Hagen–Poiseuille flow and Hooke’s Law deformation were used to confirm the numerical reliability of the underlying FEM models.

Taken together, these results provide computational evidence that capsule-based self-healing is mechanically feasible within a DIW-printed epoxy composite across all three lifecycle stages investigated — manufacturing survival, in-service activation, and healing-agent delivery. This is, to the author’s knowledge, the first study to evaluate this specific feasibility question computationally, and the first to apply machine learning as an independent cross-validation method alongside finite element simulation in this context. As a purely computational study, the conclusions presented here establish mechanical feasibility rather than demonstrated healing performance; physical fabrication of DIW-printed, capsule-embedded specimens and direct experimental measurement of healing efficiency remain necessary as the next stage of this research.

Keywords: Self-healing composites, microencapsulation, Direct Ink Writing, finite element analysis, stress concentration, capillary flow, machine learning, urea-formaldehyde microcapsules, dicyclopentadiene

TABLE OF CONTENTS

Certificate	2
Acknowledgement	3
Abstract	4-5
1. Introduction	7
1.1 Background and Motivation	7-8
1.2 Research Gap	8
1.3 Objectives of the Study	8-9
1.4 Scope of the Study	9
1.5 Literature Review	9-15
2. Methodology	16
2.1 Overall Research Approach	16
2.2 Simulation 1: DIW Extrusion Shear Stress Analysis	16-19
2.3 Simulation 2: Stress Concentration Analysis	19-24
2.4 Simulation 3: Capillary Flow Analysis	24-26
2.5 Machine Learning Cross-Validation	26-28
2.6 Statistical Analysis	28-29
3. Results and Discussion	30
3.1 Simulation 1 Results	30-32
3.2 Simulation 2 Results	32-36
3.3 Simulation 3 Results	37-38
3.4 Machine Learning Validation Results	38-40
3.5 Mesh Convergence and Statistical Analysis	40-41
4. Conclusion	42-43
5. Future Scope	44-45
References	

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Structural polymer composites are susceptible to internal damage — microcracking, delamination, and fatigue crack growth — that often occurs locations inaccessible for visual inspection or manual repair. Conventional repair approaches rely on damage being detected and externally repaired, a process that is reactive, labor-intensive, and frequently impractical for embedded or hard-to-reach structural components. This limitation motivated the development of self-healing materials: composites engineered to autonomously detect and repair damage without external intervention.

The most widely studied approach to autonomic self-healing in structural polymers is capsule-based self-healing, first demonstrated by White et al. (2001) [24]. In this approach, a healing agent — typically dicyclopentadiene (DCPD) is encapsulated within microcapsules and dispersed throughout the polymer matrix together with a catalyst. When a propagating crack ruptures a nearby capsule, the released healing agent flows into the crack and polymerizes on contact with the catalyst, sealing the damage without any external intervention. This mechanism, and the urea-formaldehyde (UF) microcapsule synthesis route developed by Brown et al. (2003) [6], have since been validated extensively, with reported fracture toughness recovery of up to 90% [4] and fracture toughness improvements of up to 127% relative to neat epoxy from capsule incorporation alone [3].

In parallel with the maturation of self-healing chemistry, additive manufacturing has emerged as an increasingly capable route for producing structural polymer composites. Direct Ink Writing (DIW), an extrusion-based 3D printing technique in which a paste-like ink is deposited layer-by-layer through a moving nozzle, has been shown to produce composites with mechanical properties approaching or exceeding those of conventionally manufactured parts, particularly when the ink exhibits shear-thinning rheological behaviour that permits extrusion under pressure while retaining shape after deposition [8]. Because DIW operates at room temperature, unlike fused deposition modelling (FDM), it is, in principle, compatible with embedding temperature-sensitive microcapsules — a possibility not available to high-temperature printing methods.

Despite the individual maturity of both capsule-based self-healing and DIW printing as separate fields, their combination — specifically, whether capsule-based self-healing agents can survive the manufacturing conditions of DIW printing and remain functional afterward — has not been systematically investigated. This project addresses that open question through a structured computational feasibility study.

1.2 Research Gap

The specific research gap motivating this project was identified from a 2019 review by Bekas, Hou, Liu and Panesar, published in *Composites Part B: Engineering* [2], which is the paper that initiated this project and to which the present work traces its origin. The review surveys the functionalities that have been incorporated into polymer-based composites via 3D printing and explicitly notes that, while additive manufacturing has matured toward production of high-value, single-function components, fabricating composites with multiple, sophisticated functionalities — including self-healing — via 3D printing remains comparatively underdeveloped [2]. This gap is independently reinforced by Jiang et al. (2023) [17], who explicitly state that the processability of self-healing materials by additive manufacturing has not been studied as thoroughly as the healing chemistries themselves, identifying this as a limiting factor for wider adoption. Sanders et al. (2019) [26] make the gap still more specific: in demonstrating successful microcapsule incorporation into stereolithography (SLA)-printed composites, the authors explicitly note that extending the same self-healing capability to Direct Ink Writing — which requires substantially more rheological optimisation of the printing ink than SLA — remained an open challenge not addressed by their own study.

Taken together, these three sources establish a clear and specific gap: it has not yet been determined, computationally or experimentally, whether capsule-based self-healing agents can survive the mechanical conditions of DIW extrusion — specifically the shear stresses generated as paste-like epoxy ink, carrying embedded microcapsules, is forced through a narrow printing nozzle under pressure. This is the gap the present study addresses.

1.3 Objectives Of The Study

The overall objective of this project is to computationally evaluate the end-to-end mechanical feasibility of capsule-based self-healing in a DIW-printed epoxy composite. This breaks down into three specific, sequential objectives, each corresponding to a mechanical prerequisite that must hold true for the self-healing mechanism to function as intended:

Objective 1: To determine whether embedded urea-formaldehyde microcapsules survive the shear stresses generated during DIW extrusion of epoxy paste through a printing nozzle, across a realistic range of printing pressures and capsule sizes (Simulation 1).

Objective 2: To determine whether embedded capsules experience sufficient stress concentration under in-service mechanical loading to rupture preferentially when a crack approaches, and to identify the capsule size and concentration that optimises this behaviour (Simulation 2).

Objective 3: To determine whether released DCPD healing agent flows into and fills a crack of realistic width within a practically useful timescale once capsule rupture has occurred (Simulation 3).

A supporting objective is to independently cross-validate the finite element results of all three simulations using machine learning regression models trained on the parametric simulation data, and to extract simplified analytical design equations relating key process and design variables to mechanical outcomes.

1.4 Scope Of The Study

This project is a purely computational feasibility study. No physical specimens were fabricated, and no experimental testing was performed; all results presented in this report are derived from finite element simulation (COMSOL Multiphysics and ANSYS Mechanical), machine learning regression models, and analytical/dimensionless-number validation. The study is limited to a single healing chemistry (DCPD healing agent encapsulated in urea-formaldehyde shells) and a single matrix material (epoxy), consistent with the most extensively validated capsule-based self-healing system reported in the literature [6], [24]. The DIW process is modelled as steady, room-temperature, single-phase laminar extrusion; curing kinetics, layer-by-layer deposition effects, and post-print mechanical processing are outside the scope of this study. The findings of this study establish computational evidence of mechanical feasibility; they do not constitute a measurement of actual healing efficiency, which would require physical fabrication and testing as a subsequent stage of research.

1.5 Literature Review

The literature reviewed for this study spans five interconnected areas: the foundational self-healing mechanism and microcapsule fabrication; Direct Ink Writing and additive manufacturing of composites; stress concentration and micromechanics around embedded

inclusions; capillary flow modelling of healing-agent transport; and the use of machine learning as a cross-validation method alongside finite element simulation. Each is reviewed in turn below.

1.5.1 Self-Healing Mechanism and Microcapsule Fabrication

The foundational work in this area is White et al. (2001) [24], who developed an epoxy matrix containing microencapsulated DCPD healing agent together with a dispersed Grubbs' ruthenium catalyst. When a propagating crack ruptures the embedded microcapsules, DCPD is released into the crack plane, and contact with the catalyst triggers ring-opening metathesis polymerization (ROMP), bonding the crack faces without external intervention. This established the core capsule-based self-healing principle adopted by nearly all subsequent research in the field, including the present study.

Brown, Kessler, Sottos and White (2003) [6] subsequently detailed the manufacturing process for the UF microcapsules used to encapsulate DCPD, produced via in situ polymerisation in an oil-in-water emulsion. By varying the agitation rate between 200 and 2000 rpm during emulsification, the authors demonstrated control of capsule diameter across a wide range — approximately 10 to 1000 μm — with a log-linear relationship observed between mean capsule diameter and agitation rate. This paper provides the literature basis for the capsule radius values (0.10 to 0.25 mm) used throughout the present study's parametric simulations.

Brown, White and Sottos (2004) [3] quantified how microcapsule size and concentration affect fracture toughness, using tapered double-cantilever beam (TDCB) specimens. Adding DCPD-filled UF microcapsules increased virgin fracture toughness by up to 127% relative to neat epoxy, substantially greater than the improvement achieved using solid filler particles of similar concentration, attributed to capsule-induced hackle markings and subsurface microcracking around the fluid-filled inclusions — an effect not observed with rigid solid particles. This supports treating embedded capsules as fluid-filled inclusions, rather than rigid fillers, in finite element modelling, an assumption carried through to the present study's stress concentration simulation.

An earlier companion study by Brown, Sottos and White (2002) [4] established the TDCB fracture testing protocol used throughout this body of work and reported that the DCPD/UF self-healing system could recover as much as 90% of virgin fracture toughness after a single autonomous healing event, depending on capsule size and concentration — a key benchmark

figure motivating why reliable capsule rupture-and-release is the mechanism that any manufacturing process, including DIW printing, must preserve.

Brown, White and Sottos (2006) [5] extended this work to fatigue loading, testing capsule concentrations of 0, 5, 10, and 20 wt% and diameters of 50, 180, and 460 μm under cyclic Mode-I loading. Microcapsule addition significantly reduced the fatigue crack growth rate and increased fatigue life, with the Paris-law exponent dropping from 9.7 for neat epoxy to approximately 4.5 at concentrations above 10 wt%, attributed to both general toughening mechanisms and crack-tip shielding from released healing fluid. This result establishes capsule concentration as a first-order design variable affecting composite mechanical performance, directly motivating the present study's capsule-concentration parametric study under Simulation 2.

Cho, White and Braun (2009) [7] broadened the application of capsule-based self-healing beyond structural fracture repair to protective coatings, demonstrating autonomic corrosion protection in damaged and healed steel panels using complementary microencapsulation approaches. Kothari and Iroh (2023) [18] more recently focused on UF microcapsule synthesis specifically in the context of composites requiring higher glass transition temperatures than many conventional self-healing chemistries can tolerate, supporting UF as a shell material compatible with structural composite applications. Yin, Rong and Zhang (2007) [33] developed an alternative two-component healing chemistry using UF microcapsules (30–70 μm diameter) filled with epoxy resin healing agent paired with a latent hardener dispersed in the matrix, demonstrating that capsule-based healing is not chemically restricted to the DCPD/Grubbs' catalyst system. Yazıcı et al. (2025) [32] developed ceramic-shell macrocapsules filled with diallyl phthalate (DAP) resin for high-temperature applications where UF capsules would be thermally unsuitable, reporting healing efficiencies of up to 69% — demonstrating that capsule shell material selection remains an active area of development for applications beyond the scope of the present study.

Two review papers provide broader context. Ilyaei, Sourki and Akbari (2021) [16] summarise capsule-based healing systems across coatings, dental composites, concrete, and polymer composites, establishing capsule properties — size, shell material, concentration — as the central design variables governing self-healing performance across this entire range of applications. Foteinidis et al. (2022) [10] demonstrate that UF-capsule-based composites can be engineered for combined mechanical and electrical (self-sensing) functionality without

degrading initial mechanical performance, broadening the argument that capsule-based self-healing is a versatile, increasingly mature platform technology — assuming, as the present study investigates, that the capsules survive the manufacturing process intact.

Two further studies address a known limitation of capsule-based healing: that it can repair only a single damage event at a given location, since local healing agent is depleted after first use. Toohey et al. (2007) [28] and Hansen et al. (2009) [14] both developed microvascular network-based alternatives, using direct-write assembly to embed three-dimensional channel networks that can be repeatedly replenished, demonstrating autonomous repair of repeated damage at the same location over at least 30 cycles — a capability inherently unavailable to the capsule-based mechanism investigated in the present study. This trade-off is acknowledged explicitly in the discussion and future scope of this report.

1.5.2 Direct Ink Writing and Additive Manufacturing of Composites

Compton and Lewis (2014) [8] reported an epoxy-based ink developed specifically for DIW of lightweight cellular composites, achieving Young’s modulus values up to ten times higher than commercially available 3D-printed polymers. Critically, the ink’s shear-thinning rheological behaviour is what permits extrusion through a narrow nozzle under moderate pressure while retaining filament shape after deposition — this paper is a foundational reference establishing epoxy-based DIW as a viable manufacturing route, and provides literature precedent for the shear-thinning viscosity assumption used in the present study’s Simulation 1.

Lewicki et al. (2017) [20] reported the first additively manufactured carbon-fibre-reinforced composite produced via DIW, using a latent thermally cured resin system with controlled three-dimensional fibre alignment achieved through micro-extrusion; notably, the development process included computational flow modelling to inform the extrusion process design, directly preceding the present study’s use of COMSOL-based flow simulation. Hansen et al. (2009) [14], in addition to its microvascular network contribution discussed above, reports nozzle diameters in the 100–500 μm range for direct-write assembly — the same general scale as the 0.4 mm nozzle radius modelled in the present study’s Simulation 1.

Gonzalez et al. (2023) [13] investigated UV-assisted hybrid DIW of fibre-reinforced thermoset composites, addressing DIW’s typical limitations in achievable print height and design complexity through in-process UV curing. Wang et al. (2025) [29] addressed incomplete curing and weak interfacial bonding in DIW of continuous-fibre composites

through a pre-impregnation strategy, reinforcing that resin viscosity and curing-condition optimisation remain active, unresolved engineering challenges in DIW processing. Sanders et al. (2019) [26], as noted above, demonstrated successful capsule incorporation into SLA-printed composites while explicitly flagging DIW's greater rheological optimisation burden as an open challenge — the single most directly relevant literature statement motivating the present study's research question.

Two broader review papers situate DIW within the wider additive manufacturing landscape for composites. Safari, Kami and Abedini (2022) [25] review fused deposition modelling of continuous fibre-reinforced composites, establishing that systematic process-parameter studies — of the kind undertaken in the present study's pressure and capsule-size parametric analyses — are a standard and expected methodology across 3D-printing composite research generally. Nadgorny and Ameli (2018) [23] review functional and smart polymer 3D printing more broadly, concluding that materials research, rather than printing technology itself, remains the critical bottleneck for realising functional 3D-printed structures — directly supporting the present study's framing of material-process compatibility as the central open question.

Wu, Deconinck and Lewis (2011) [30] used omnidirectional (direct-write) printing of fugitive ink to pattern arbitrary three-dimensional microvascular networks, demonstrating that extrusion-based printing techniques are broadly compatible with healing-relevant architectures, while reinforcing that ink and matrix rheological compatibility is the central technical constraint — a recurring theme across this literature directly motivating the present study's detailed simulation of flow and shear stress during DIW extrusion.

1.5.3 Stress Concentration and Micromechanics of Embedded Inclusions

Ahmed and Sanada (2019) [1] developed a micromechanical finite element model to predict the effective elastic properties of capsule-based composites, using dynamic mechanical analysis to characterize matrix modulus and single-microcapsule micromanipulation testing combined with 2D FEM to determine capsule modulus, before building detailed 3D FEM models examining different capsule packing arrangements and volume fractions. This is the closest direct precedent for the present study's approach of modelling capsules as distinct mechanical inclusions within the epoxy matrix using finite element analysis.

Gao et al. (2021) [11] developed mesoscale finite element models of polymer matrix composites containing self-healing microcapsules, cross-referenced against experimental data to investigate the stress states and microstructural parameters governing the self-healing

mechanism. This is the closest literature analogue to the present study's Simulation 2, directly supporting the modelling approach of treating capsules as distinct domains within the matrix and evaluating stress concentration via FEM.

Kushch (2003) [19] developed a semi-analytical multipole-expansion solution for stress concentration around spherical inclusions in a transversely isotropic elastic matrix, providing a rigorous analytical benchmark independent of finite element computation. McDonald et al. (2019) [22] used X-ray computed tomography and phase-contrast imaging to directly visualise, in three dimensions, which embedded microcapsules actually rupture as a crack propagates — offering an experimental basis confirming that capsules near a crack preferentially rupture, the underlying premise of the present study's stress concentration analysis.

1.5.4 Capillary Flow Modelling of Healing-Agent Transport

The foundational reference for this area is Washburn (1921) [31], who derived what is now known as the Lucas–Washburn equation, describing capillary flow of a liquid into a narrow channel as a balance of surface-tension driving force against viscous resistance, predicting that penetration distance increases with the square root of time. This relationship is the governing equation directly implemented as a time-dependent ODE in the present study's Simulation 3.

Gardner et al. (2014) [12] presented experimental and numerical simulations of capillary flow of a healing agent through discrete cracks in cementitious materials, extending the basic Washburn relationship with corrections for stick-slip meniscus behaviour and frictional dissipation, validated against two independent experimental benchmarks. This is the closest direct literature analogue to the present study's Simulation 3, confirming the basic Washburn-type equation as an accepted starting point while indicating that the present study's simpler, uncorrected model represents a first-pass simplification relative to the current state of the art. Yip et al. (2025) [34] more recently developed a dynamic contact angle correction within a volume-of-fluid framework, explicitly presented as an improvement over the Gardner et al. approach for complex, non-uniform crack geometries — a useful point of reference when discussing the limitations of the present study's simplified geometry.

1.5.5 Machine Learning as a Cross-Validation Method for Finite Element Simulation

Hashemi, Jang and Beheshti (2023) [15] developed machine learning-based surrogate models — including decision trees, random forest, gradient boosting, and deep neural networks — trained on finite element analysis data to estimate the dynamic response of mechanical systems as a computationally faster alternative to repeated FEA. This is the direct

methodological precedent for the present study's use of machine learning as an independent cross-validation method for the COMSOL and ANSYS results, rather than as a replacement for simulation.

Mahajan et al. (2024) [21] used Random Forest regression, further optimised via genetic-algorithm hyperparameter tuning, to predict the flexural properties of composites from manufacturing and design variables, providing a close methodological analogue for the present study's Random Forest model structure. Sharma et al. (2025) [27] review machine learning methods — including artificial neural networks, support vector machines, random forests, and Gaussian process regression — applied to composite mechanical property prediction across 2020–2025 literature, supporting Random Forest regression as a well-precedented algorithm choice for the present dataset.

Finally, Zhelyazov (2026) [35] used finite element analysis to simulate the quasi-static mechanical response of concrete specimens containing self-healing microcapsules, evaluating both the damage-inducing preload phase and subsequent property recovery. Although applied to a cementitious rather than epoxy matrix, this is a close and recent methodological analogue to the present study's overall computational approach, reinforcing that FEM-based simulation is an accepted and active method for evaluating capsule-based self-healing performance across different host matrix materials.

CHAPTER 2

METHODOLOGY

2.1 Overall Research Approach and Manufacturing Process Selection

The overall research approach follows directly from the three objectives stated in Section 1.3. The review’s classification of 3D printing methods by enabling technology [2] provided the basis for selecting which printing method is mechanically compatible with embedding pre-formed microcapsules. Methods involving high process temperature, such as fused deposition modelling, or photopolymer chemistry, such as stereolithography, were excluded as candidates on the grounds that they would either thermally degrade the UF shell and DCPD healing agent, or chemically interfere with the capsule shell or healing-agent curing mechanism.

The shear-thinning rheological behaviour of epoxy-based DIW inks reported by Compton and Lewis [8] is the basis for selecting Direct Ink Writing as the manufacturing process under simulation, since this behaviour allows paste-like ink to be extruded through a narrow nozzle under pressure while retaining its shape after deposition, at room temperature, avoiding thermal degradation of embedded capsules. The explicit observation by Sanders et al. [26] that DIW requires substantially more rheological optimisation than SLA is the basis for dedicating a full simulation — Simulation 1 — specifically to characterising flow behaviour and shear stress during DIW extrusion, rather than treating the manufacturing step as a given.

The overall methodology therefore comprises three sequential finite element simulations, each addressing one objective from Section 1.3, supported by a machine learning cross-validation stage and a statistical analysis stage.

2.2 Simulation 1 — DIW Extrusion Shear Stress Analysis (COMSOL Multiphysics)

2.2.1 Governing Equations and Physics

The epoxy paste was modelled as an incompressible viscous fluid under laminar flow conditions, governed by the steady-state Navier–Stokes and continuity equations:

$$\rho(\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0$$

At the operating Reynolds number ($Re = 0.0096$, Table 4), inertial terms are negligible relative to viscous terms, reducing the governing equation to the Stokes flow limit, $\mu \nabla^2 \mathbf{u} = \nabla p$.

The COMSOL Multiphysics Laminar Flow interface (spf) was used to solve this system over a 2D axisymmetric domain.

2.2.2 Geometry, Materials, and Boundary Conditions

The simulation domain represents a single DIW nozzle in 2D axisymmetric form, with one spherical urea-formaldehyde capsule positioned within the flow near the nozzle wall, where shear is expected to be highest.

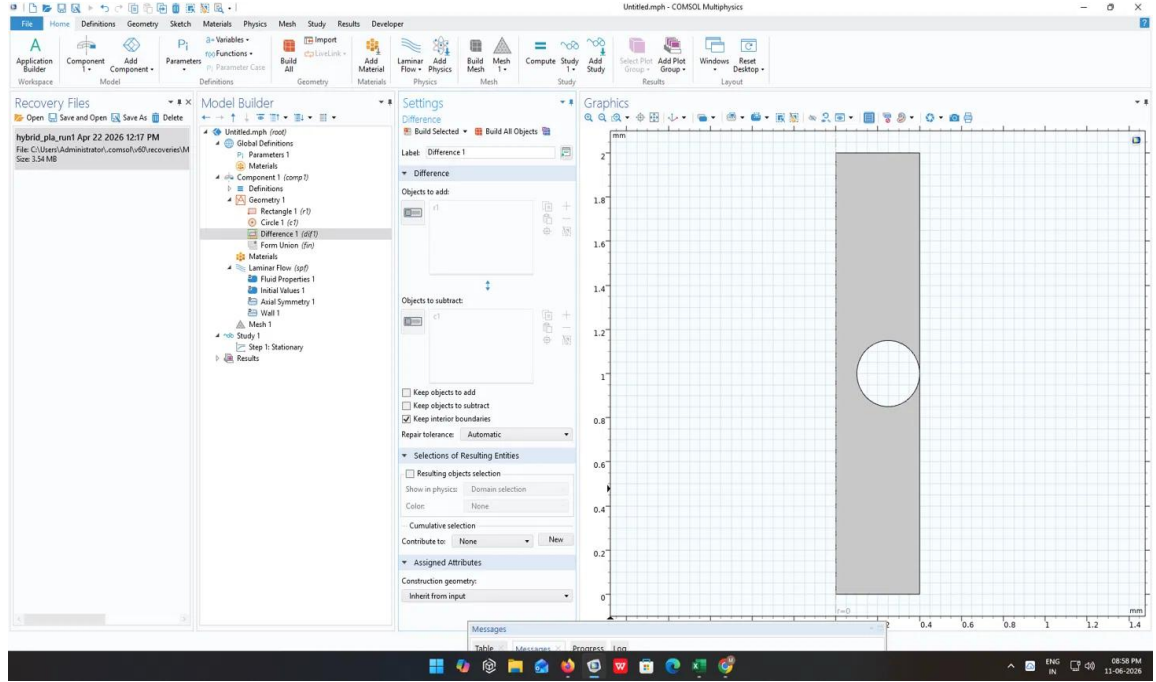


Figure 1: 2D axisymmetric geometry of the DIW nozzle with embedded capsule, as built in COMSOL Multiphysics. Nozzle inner radius = 0.4 mm, nozzle length = 2 mm, capsule radius = 0.15 mm (baseline case), capsule centred at $r = 0.25$ mm, $z = 1$ mm.

Table 1: Material Properties Used in Simulation 1

Parameter	Value	Source / Justification
Epoxy paste density (ρ)	1200 kg/m ³	Literature DIW epoxy inks [8]
Epoxy paste dynamic viscosity (μ)	10 Pa·s	Shear-thinning epoxy paste at print speed
UF capsule shell density	1400 kg/m ³	Brown et al. 2003 [6]
UF capsule shell Young's modulus	3700 MPa	Literature UF shell properties
UF capsule shell Poisson's ratio	0.30	Literature UF shell properties
UF shell failure strength	≈40 MPa	Literature benchmark for UF shell rupture
Nozzle inner radius	0.4 mm	Hansen et al. 2009 [14]
Nozzle length	2 mm	Simulation geometry

Table 2: Boundary Conditions — Simulation 1 (DIW Nozzle Flow)

Boundary	Condition Type	Value
Inlet (top face, $z = 2$ mm)	Pressure inlet	Variable: 0.5, 1.0, 1.5, 2.0, 2.5, 3.0 bar
Outlet (bottom face, $z = 0$)	Pressure outlet	0 Pa (atmospheric reference)
Nozzle wall (outer radius)	No-slip wall	$u = 0$ m/s
Capsule surface	No-slip rigid wall	$u = 0$ m/s (capsule treated as rigid)
Symmetry axis ($r = 0$)	Axial symmetry	2D axisymmetric assumption

2.2.4 Parametric Study Design

Two parametric studies were carried out. The first varied inlet pressure from 0.5 to 3.0 bar at fixed capsule radius (0.15 mm), reflecting the realistic operating range for DIW printing. The second varied capsule radius from 0.10 to 0.25 mm at fixed inlet pressure (1.0 bar), reflecting the achievable capsule size range established by Brown et al. [6]. This design was motivated by the finding of Brown et al. (2006) [5] that capsule concentration and size are first-order variables affecting composite mechanical performance, supporting the decision to vary capsule size systematically rather than treat it as fixed.

2.2.5 Validation Approach

The COMSOL flow field was validated against the analytical Hagen–Poiseuille solution for centreline velocity in an empty cylindrical pipe:

$$V_{\max} = (R^2 \times \Delta P) / (4 \times \mu \times L) = (0.0004^2 \times 100000) / (4 \times 10 \times 0.002) = 2 \times 10^{-4} \text{ m/s} = 0.2 \text{ mm/s}$$

The COMSOL simulation, with the capsule present, gave a maximum velocity of approximately 0.1 m/s — a reduction consistent with the 37.5% blockage ratio introduced by the capsule occupying part of the nozzle cross-section (Table 4), confirming the result is physically reasonable rather than numerically erroneous.

Table 4: Dimensionless Number Validation — Simulation 1

Dimensionless Number	Formula	Value	Interpretation
Reynolds Number (Re)	$\rho V D / \mu$	0.0096	$Re \ll 2300 \rightarrow$ laminar flow confirmed
Capillary Number (Ca)	$\mu V / \gamma$	32.25	$Ca \gg 1 \rightarrow$ viscous forces dominate; single-phase assumption valid
Stokes Number (St)	$\rho V D_{\text{cap}} / \mu$	0.0036	$St \ll 1 \rightarrow$ capsule follows fluid flow; rigid-wall treatment valid
Blockage Ratio	$D_{\text{cap}} / D_{\text{nozzle}} \times 100$	37.5%	Explains ~50% velocity reduction vs. empty pipe

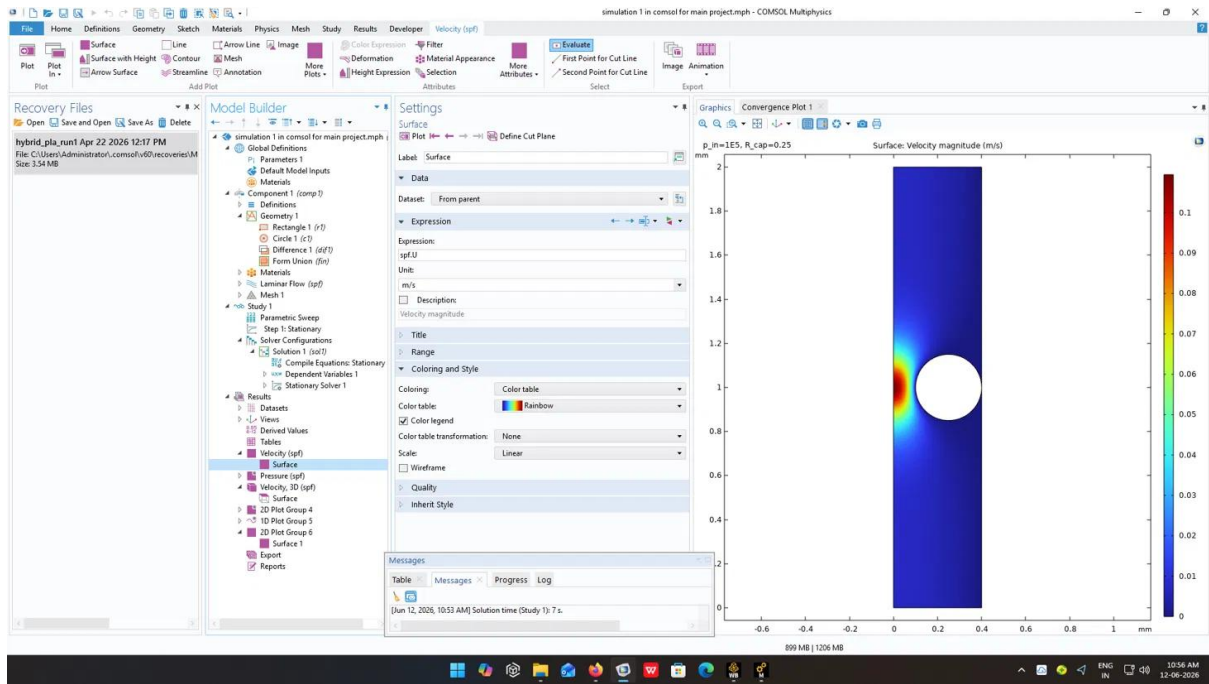


Figure 2: COMSOL velocity magnitude plot (spf.U) for Simulation 1 at 1 bar inlet pressure, showing the velocity field around the embedded capsule.

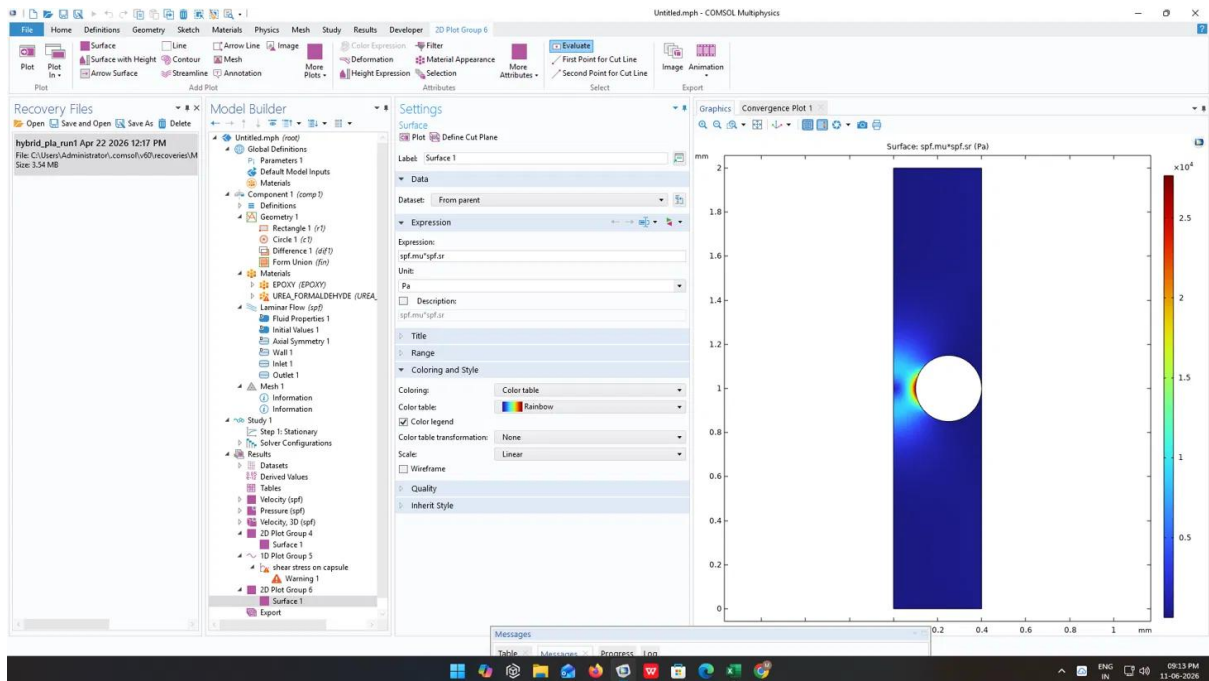


Figure 3: COMSOL shear stress distribution (spf.mu × spf.sr) for Simulation 1, showing maximum shear stress concentrated at the capsule's leading and trailing edges.

2.3 Simulation 2 — Stress Concentration Analysis (ANSYS Mechanical)

2.3.1 Governing Approach: Capsules as Mechanical Inclusions

Simulation 2 evaluates the second mechanical prerequisite — that capsules must rupture preferentially when a crack approaches during service. The ANSYS finite element model treats

capsules as distinct elastic inclusions within the epoxy matrix, following the mesoscale FEM approach preceded by Ahmed and Sanada [1] and Gao et al. [11]. Capsule and matrix domains were created via Boolean subtract operations in ANSYS DesignModeler, ensuring shared node connectivity at the capsule–matrix interface without separate contact elements, allowing direct evaluation of the stress state inside and around each capsule under applied loading.

2.3.2 Geometry, Materials, and Boundary Conditions

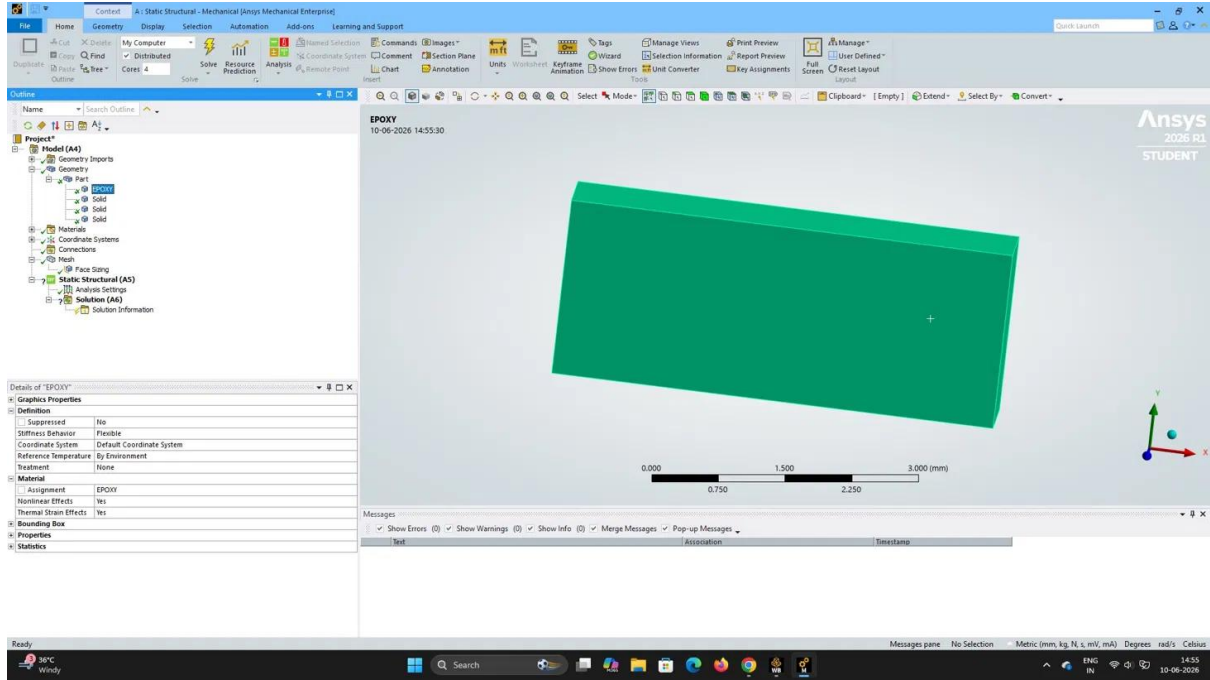


Figure 4: 3D epoxy block geometry (5 mm × 2 mm × 1 mm) with three embedded spherical capsules (radius 0.15 mm) as built in ANSYS DesignModeler.

Table 5: Material Properties — Simulation 2

Material	Property	Value	Source
Epoxy matrix	Young's Modulus (E)	3400 MPa	Standard structural epoxy
Epoxy matrix	Poisson's Ratio (ν)	0.35	Standard structural epoxy
Epoxy matrix	Density (ρ)	1200 kg/m ³	Standard structural epoxy
UF capsule shell	Young's Modulus (E)	3700 MPa	UF shell literature
UF capsule shell	Poisson's Ratio (ν)	0.30	UF shell literature
UF capsule shell	Density (ρ)	1400 kg/m ³	Brown et al. 2003 [6]

Table 6: Boundary Conditions — Simulation 2

Boundary	Condition Type	Value
Left face (X = 0)	Fixed Support	All DOF constrained

Boundary	Condition Type	Value
Right face (X = 5 mm)	Tensile Force	100 N in X-direction
Capsule–matrix interface	Bonded contact	Shared nodes (Boolean subtract)
All other faces	Free surface	No constraint

2.3.3 Mesh and Mesh Convergence Study

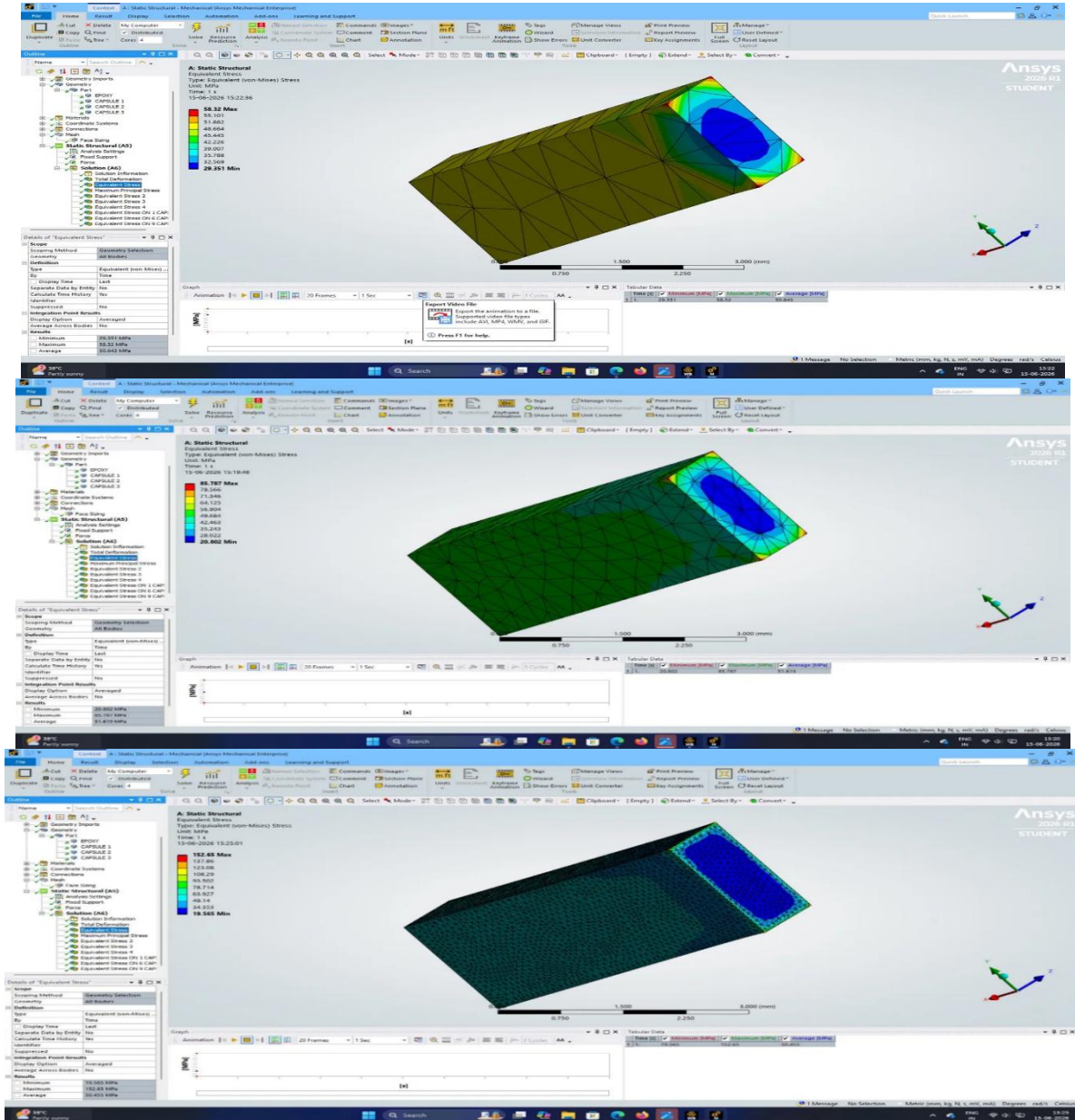


Figure 5: Mesh used in Simulation 2 at three element sizes — coarse (0.5 mm), medium (0.2 mm), and fine (0.05 mm) — showing refinement near capsule surfaces.

A mesh convergence study was performed to confirm that the results are not dependent on element size. Three mesh densities were tested, and both maximum and average Von Mises stress were recorded at each (Table 7).

Table 7: Mesh Convergence Study Results — Simulation 2

Mesh Size	Approx. Elements	Max Von Mises Stress (MPa)	Avg Von Mises Stress (MPa)	Avg Stress Error vs. Fine (%)
Coarse — 0.5 mm	~500	58.32	50.643	0.37%
Medium — 0.2 mm	~2000	85.787	51.619	2.31%
Fine — 0.05 mm	~8000	152.65	50.455	0.00% (reference)

Average Von Mises stress converges to within 2.31% across all three mesh densities, confirming the result is mesh independent. Maximum stress increases with mesh refinement due to a stress singularity at the sharp capsule–epoxy interface — a well-documented finite element phenomenon at geometric discontinuities — and is therefore not used as the convergence metric. On this basis, the medium mesh (0.2 mm) was selected for all subsequent parametric studies, balancing accuracy against computational cost.

2.3.4 Parametric Studies: Capsule Size and Concentration

Two parametric studies were carried out, motivated by the finding of Brown et al. (2006) [5] that capsule concentration systematically affects fatigue and mechanical performance, and by Kushch's [19] analytical treatment of stress concentration around spherical inclusions, which provides an independent theoretical basis for comparison with the FEM-derived stress concentration factor.

Table 8: Parametric Study Matrix — Simulation 2

Study	Variable	Values Tested	Fixed Parameter
Capsule Size Study	Capsule Radius	0.10, 0.15, 0.20, 0.25 mm	Number of capsules = 3
Concentration Study	Number of Capsules	1, 3, 6, 9 capsules	Capsule radius = 0.15 mm

2.3.5 Validation Approach

The ANSYS deformation result was validated against the analytical Hooke's Law prediction for a uniaxially loaded bar:

$$\delta = FL / (EA) = (100 \times 0.005) / (3400 \times 10^6 \times 0.002) = 7.35 \times 10^{-5} \text{ m} = 0.0735 \text{ mm}$$

The ANSYS result gave $\delta = 0.072785 \text{ mm}$, an error of 0.97% relative to the analytical prediction — confirming the finite element model is correctly configured.

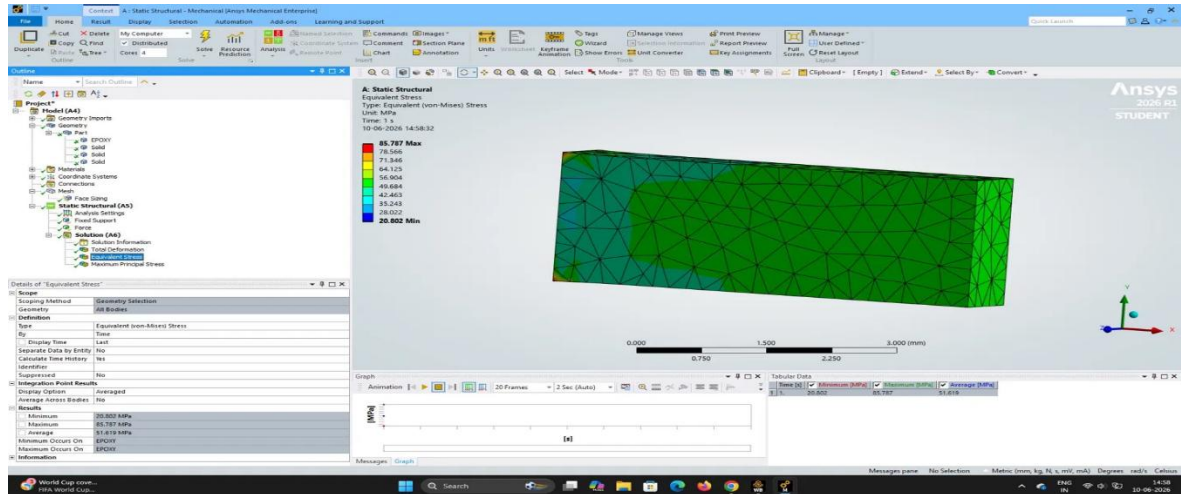


Figure 6: ANSYS Von Mises stress contour for the baseline case (3 capsules, 0.15 mm radius, 100 N), showing maximum stress 85.787 MPa concentrated at capsule surfaces.

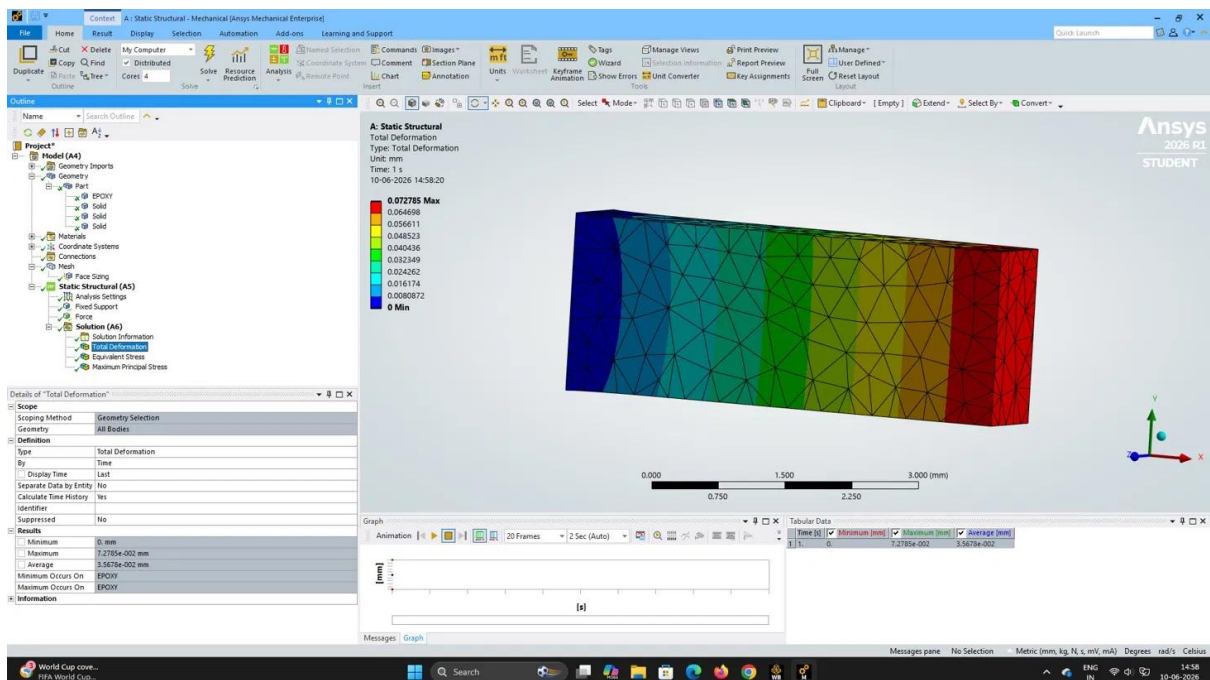


Figure 7: ANSYS total deformation contour, showing maximum deformation 0.072785 mm at the loaded face, consistent with the Hooke's Law validation above.

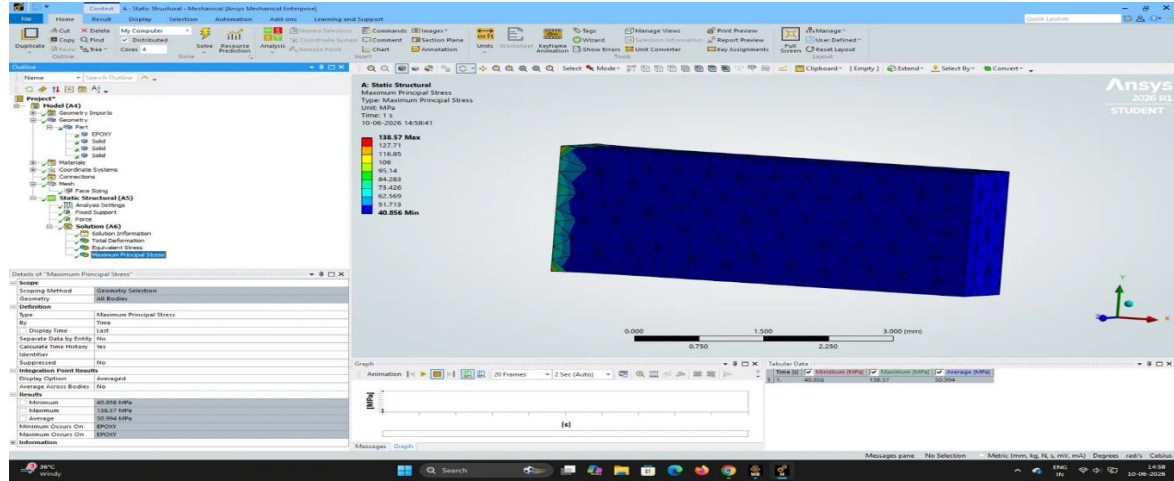


Figure 8: ANSYS maximum principal stress contour, showing maximum 138.57 MPa at capsule surface edges facing the applied tensile direction.

2.4 Simulation 3 — Healing Agent Capillary Flow (COMSOL — 1D Washburn ODE Model)

2.4.1 Governing Equation

Simulation 3 evaluates the third mechanical prerequisite — that released healing agent must flow into and fill a crack. The governing equation is the Washburn capillary flow equation [31], describing the penetration of a wetting liquid into a narrow channel under surface-tension driving force, balanced against viscous resistance:

$$dL/dt = (D \times \gamma \times \cos\theta) / (8 \times \eta \times L)$$

where L is penetration length (m), D is crack width (m), γ is surface tension (N/m), θ is contact angle ($^\circ$), and η is the dynamic viscosity of the healing agent (Pa·s). This was implemented as a time-dependent ordinary differential equation in COMSOL's Global ODEs and DAEs (ge) interface.

An initial attempt was made to model this process using the full two-dimensional Two-Phase Flow, Phase Field interface in COMSOL. This approach encountered fundamental timestep convergence difficulties at the micron length scale, with the solver timestep reducing to the nanosecond range and requiring on the order of millions of steps to reach the target simulation time of 0.05 seconds — computationally intractable on the hardware available for this study. The one-dimensional Washburn ODE formulation was therefore adopted instead. This simplification is consistent with the approach validated by Gardner et al. (2014) [12], who note that basic Washburn-type models are an accepted and standard starting point for healing-agent transport simulation, while more advanced corrections — such as the dynamic contact

angle approach of Yip et al. (2025) [34] — represent refinements beyond this baseline model that were considered outside the scope of the present study (see Chapter 5, Future Scope).

2.4.2 Material Parameters and Geometry

Table 9: Material Parameters Used in Simulation 3

Parameter	Symbol	Value	Source
Surface tension of DCPD	γ	0.031 N/m	DCPD fluid property literature
Contact angle (DCPD on epoxy)	θ	20°	Estimated partial-wetting condition
Dynamic viscosity of DCPD	η	0.00097 Pa·s	DCPD fluid property literature
Crack width (parametric)	D	5, 10, 20, 50 μm	Realistic crack width range
Simulation time	t	0 to 0.05 s	Sufficient to observe complete crack filling
Initial penetration length	L_0	1×10^{-10} m	Non-zero initial condition to avoid singularity at $t = 0$

2.4.3 Parametric Study: Crack Width

Crack width was varied across four values — 5, 10, 20, and 50 μm — to represent the realistic range of crack widths encountered in damaged epoxy composites, with penetration length recorded as a function of time for each case.

2.4.4 Validation Approach

Table 11: Dimensionless Number Validation — Simulation 3

Dimensionless Number	Formula	Value (D = 10 μm)	Interpretation
Bond Number (Bo)	$\rho g D^2 / \gamma$	3.10×10^{-5}	$\text{Bo} \ll 1 \rightarrow$ gravity negligible vs. surface tension
Ohnesorge Number (Oh)	$\mu / \sqrt{\rho \cdot \gamma \cdot D}$	573	$\text{Oh} > 1 \rightarrow$ viscous forces dominate inertia; laminar capillary flow valid

The numerically obtained penetration length at $t = 0.025$ s (1.97 mm) was compared against the closed-form Washburn analytical solution evaluated at the same time (≈ 1.39 mm); the qualitative square-root-of-time growth shape matches in both cases, with the numerical magnitude difference attributable to the non-zero initial condition required to avoid a singularity at $t = 0$ in the numerical ODE formulation.

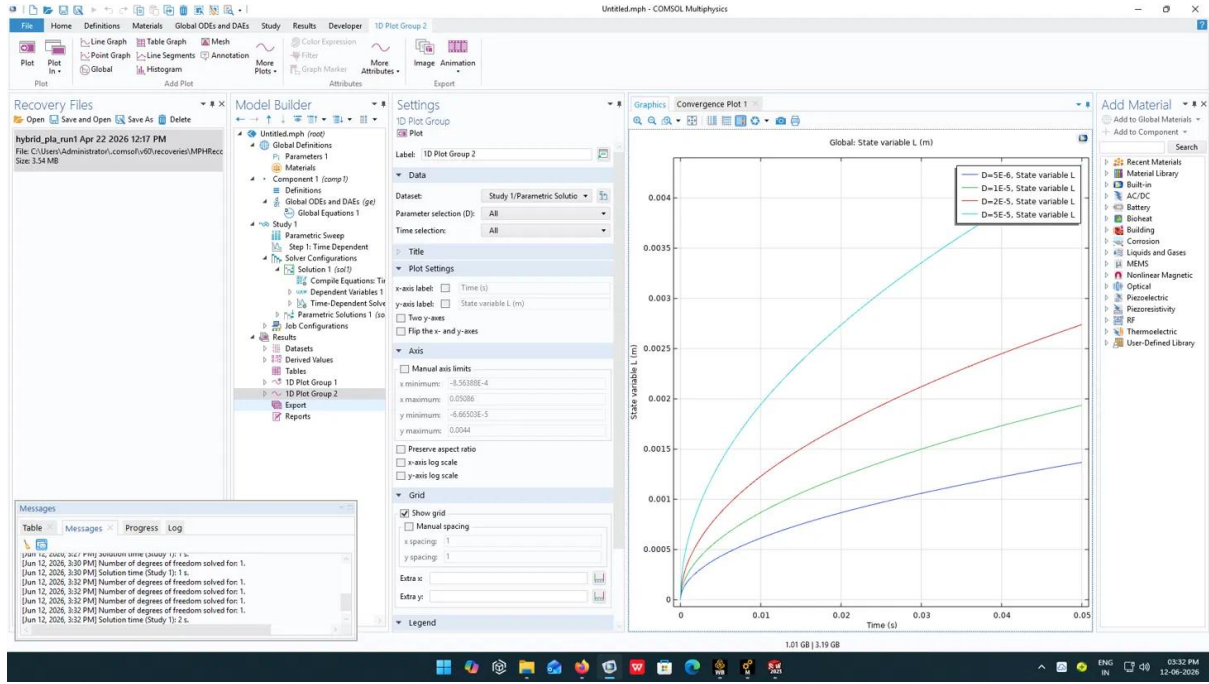


Figure 9: COMSOL penetration length vs. time curves for all four crack widths tested (5, 10, 20, 50 μm), showing square-root-of-time growth consistent with Washburn theory.

2.5 Machine Learning Cross-Validation Approach

2.5.1 Rationale

Machine learning was used as an independent cross-validation method for the finite element results, following the precedent of Hashemi et al. (2023) [15], who demonstrated that ML-based surrogate models trained on FEA data can closely reproduce FEA-predicted response, and Mahajan et al. (2024) [21] and Sharma et al. (2025) [27], who establish Random Forest regression as a well-precedented method for composite property prediction.

2.5.2 ML Model Structure and Training Approach

Table 12: ML Model Input/Output Structure

ML Model	Input Features	Output Target	Algorithm	Training Data Source
Model 1	Inlet pressure, capsule radius	Max shear stress on capsule (MPa)	Random Forest (100 trees)	10 points — Simulation 1
Model 2a	Capsule radius, no. of capsules	Max Von Mises stress (MPa)	Random Forest (100 trees)	7 points — Simulation 2
Model 2b	Capsule radius, no. of capsules	Stress Concentration Factor	Random Forest (100 trees)	7 points — Simulation 2
Model 3	Crack width	Penetration length (mm)	Polynomial Regression (deg. 2)	4 points — Simulation 3

All ML models were trained using the parametric simulation results as training data; no external dataset was used. The scikit-learn library in Python was used in a Jupyter Notebook

environment. Random Forest models used 100 estimators with a fixed random seed (42) for reproducibility. Given the small dataset sizes (4–10 points per study), no train/test split was applied; model fit quality was assessed via R^2 on the full training set, consistent with the cross-validation (rather than predictive-deployment) purpose of this analysis.

2.5.3 Design Equation Extraction

In addition to Random Forest regression, polynomial regression was used to extract simplified closed-form design equations relating key input variables to mechanical outcomes, providing equations used directly by a designer without re-running simulation.

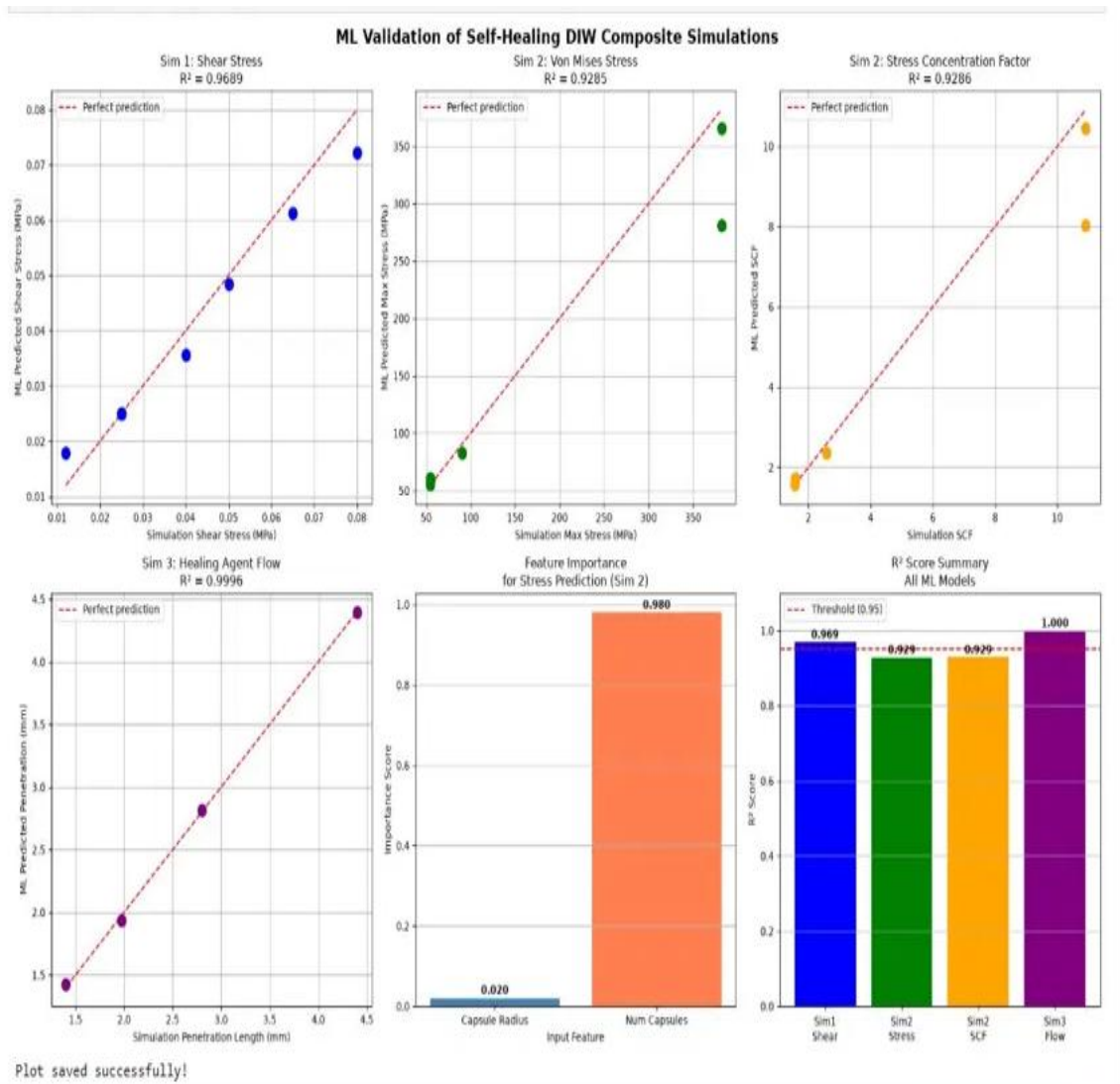


Figure 10: ML validation plots: predicted-vs-actual scatter plots for all four ML models, feature importance bar chart, and R^2 summary bar chart (six-panel figure).

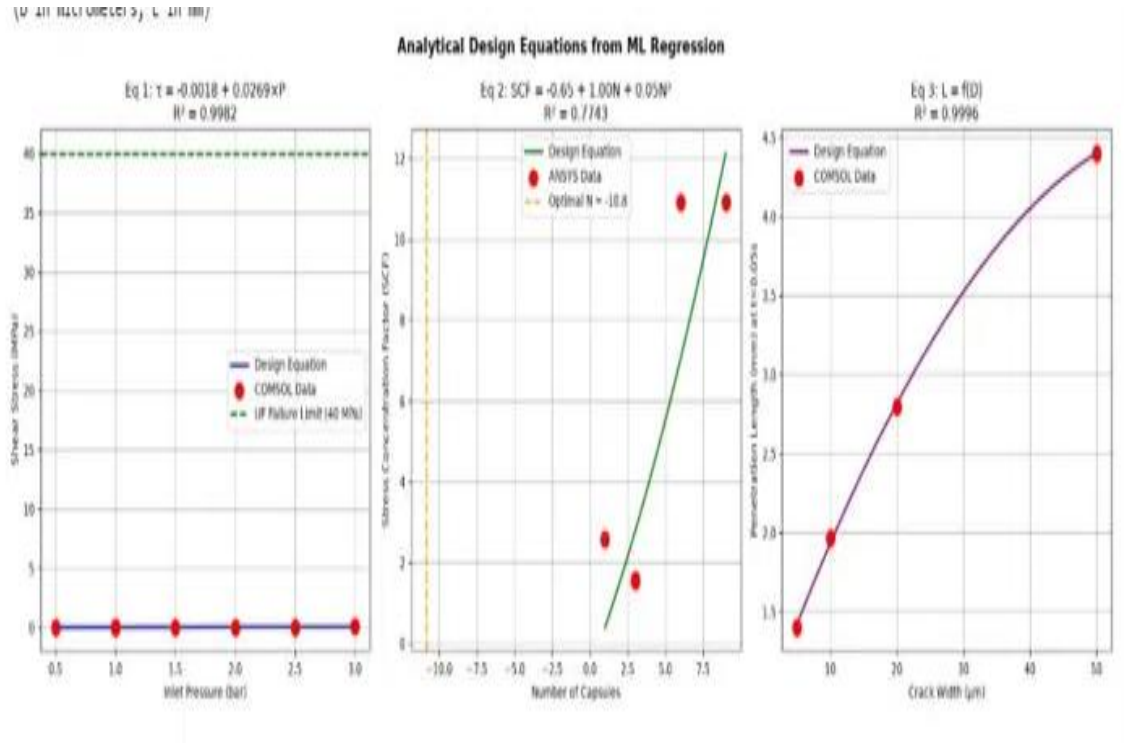


Figure 11: Design equation plots: shear stress vs. pressure, SCF vs. number of capsules, and penetration length vs. crack width, each with fitted regression curve (three-panel figure).

2.6 Statistical Analysis Approach

Statistical analysis was performed on all parametric study outputs to quantify the relative influence of each design variable. For each dataset, the arithmetic mean, standard deviation, coefficient of variation ($CV\% = \text{standard deviation} / \text{mean} \times 100$), and 95% confidence interval were calculated.

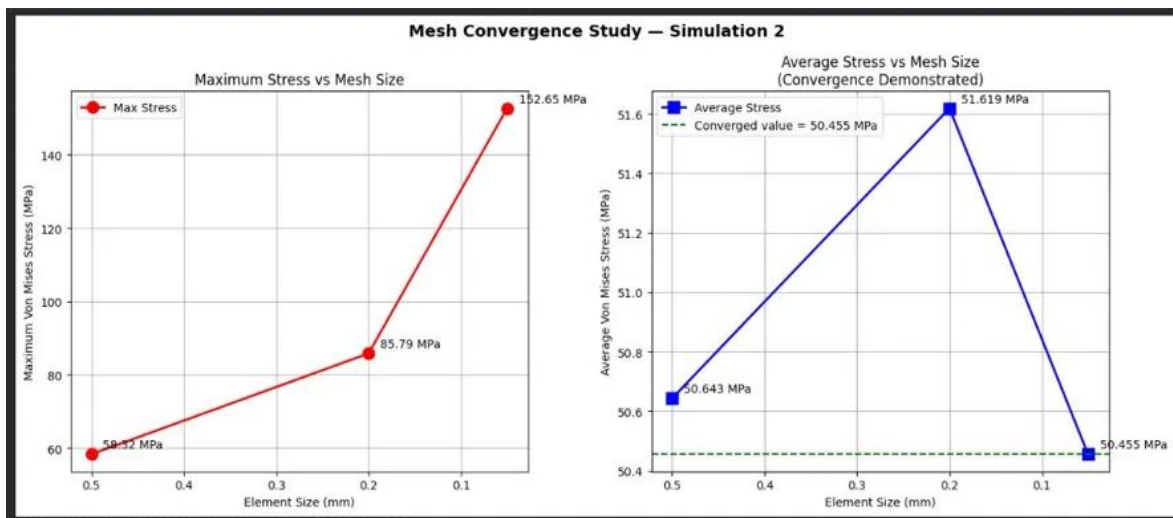


Figure 12: Mesh convergence plots: maximum stress vs. mesh size (showing the stress-singularity trend) and average stress vs. mesh size (showing convergence within 2.31%), two-panel figure

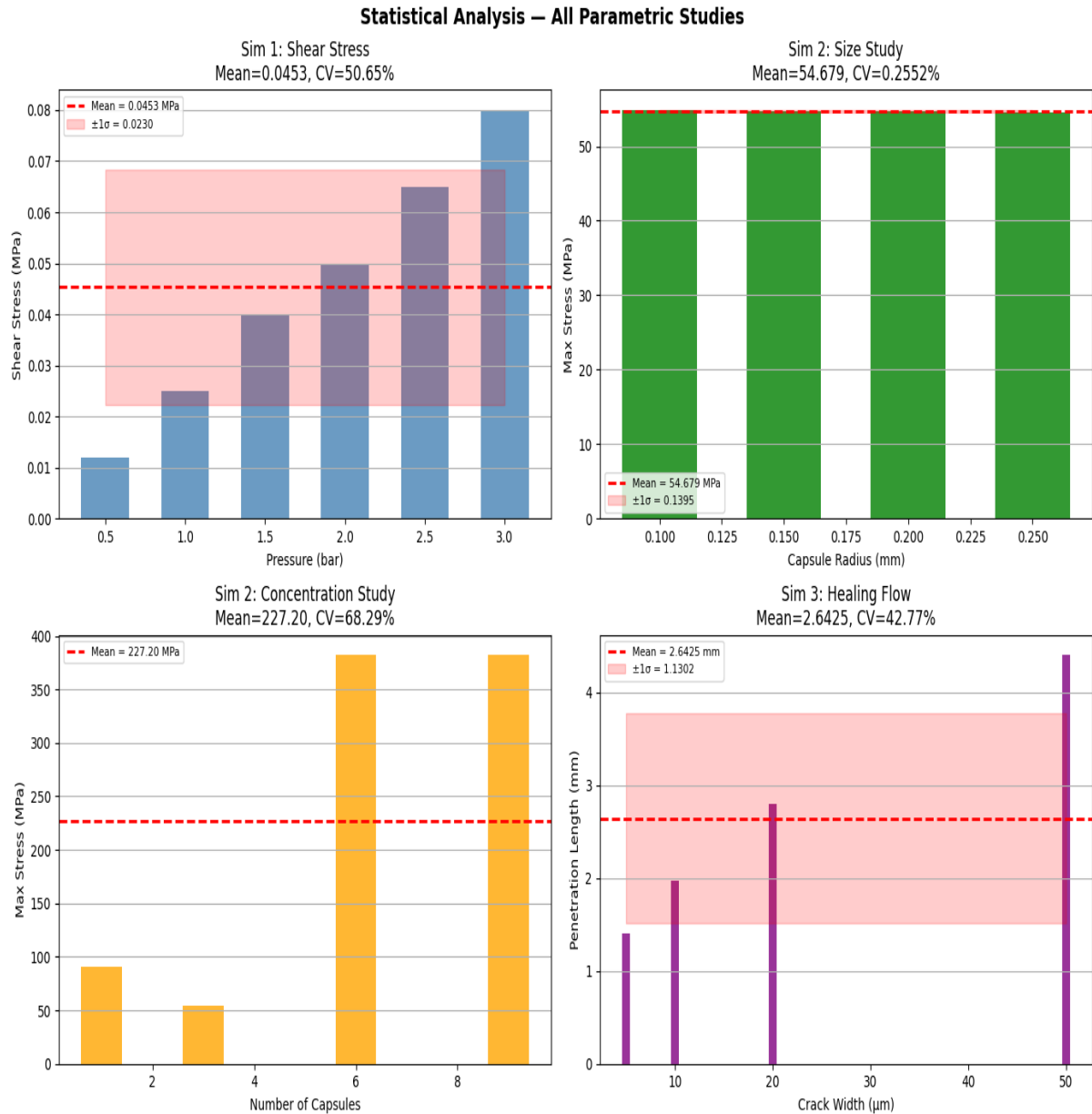


Figure 13: Statistical analysis plots: bar charts with mean and $\pm 1\sigma$ band for each parametric study — shear stress vs. pressure, stress vs. capsule size, stress vs. concentration, and penetration vs. crack width (four-panel figure).

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Simulation 1 — DIW Shear Stress Results

3.1.1 Pressure Parametric Study

Table 16: Shear Stress vs. Inlet Pressure — Safety Factor Analysis

Inlet Pressure (bar)	Max Shear Stress (MPa)	UF Shell Strength (MPa)	Safety Factor	Status
0.5	0.012	40	3333×	SAFE
1.0	0.025	40	1600×	SAFE
1.5	0.040	40	1000×	SAFE
2.0	0.050	40	800×	SAFE
2.5	0.065	40	615×	SAFE
3.0	0.080	40	500×	SAFE

Across the full range of inlet pressures tested, maximum shear stress on the capsule surface increases approximately linearly with pressure, from 0.012 MPa at 0.5 bar to 0.080 MPa at 3.0 bar — the highest pressure realistically expected in DIW printing. Even at this worst-case condition, the resulting shear stress remains approximately 500 times below the literature-reported UF shell failure strength of 40 MPa [6]. This result directly supports the conclusion that capsules survive the manufacturing process: the toughening benefit reported by Brown et al. (2004) [3], which depends on capsules reaching the cured composite intact, is not put at risk by shear forces during DIW extrusion at any pressure within the tested range.

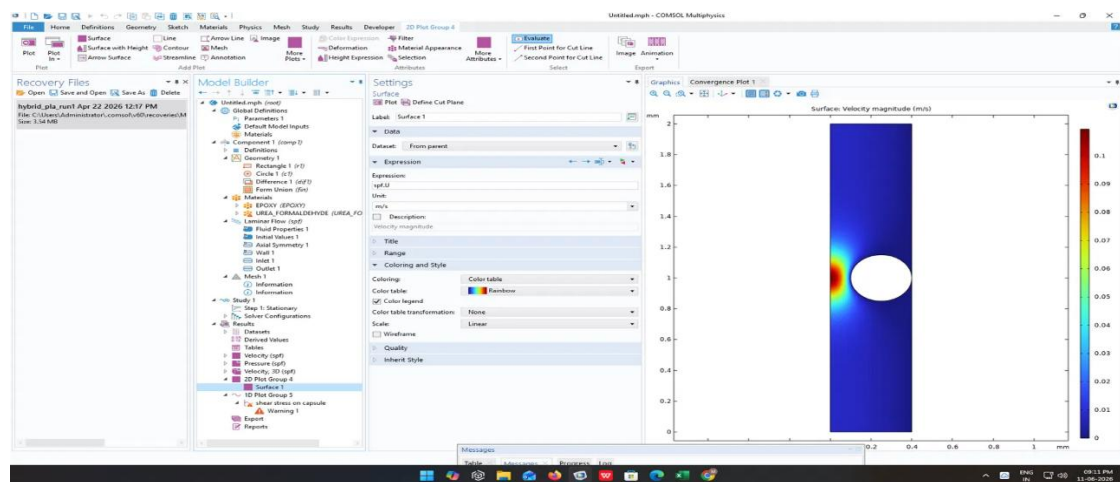


Figure 14: COMSOL velocity profile for Simulation 1 at 1 bar inlet pressure, with the Hagen–Poiseuille analytical centreline velocity (0.2 m/s, empty pipe) and the COMSOL result (≈ 0.1 m/s, with capsule) annotated for comparison.

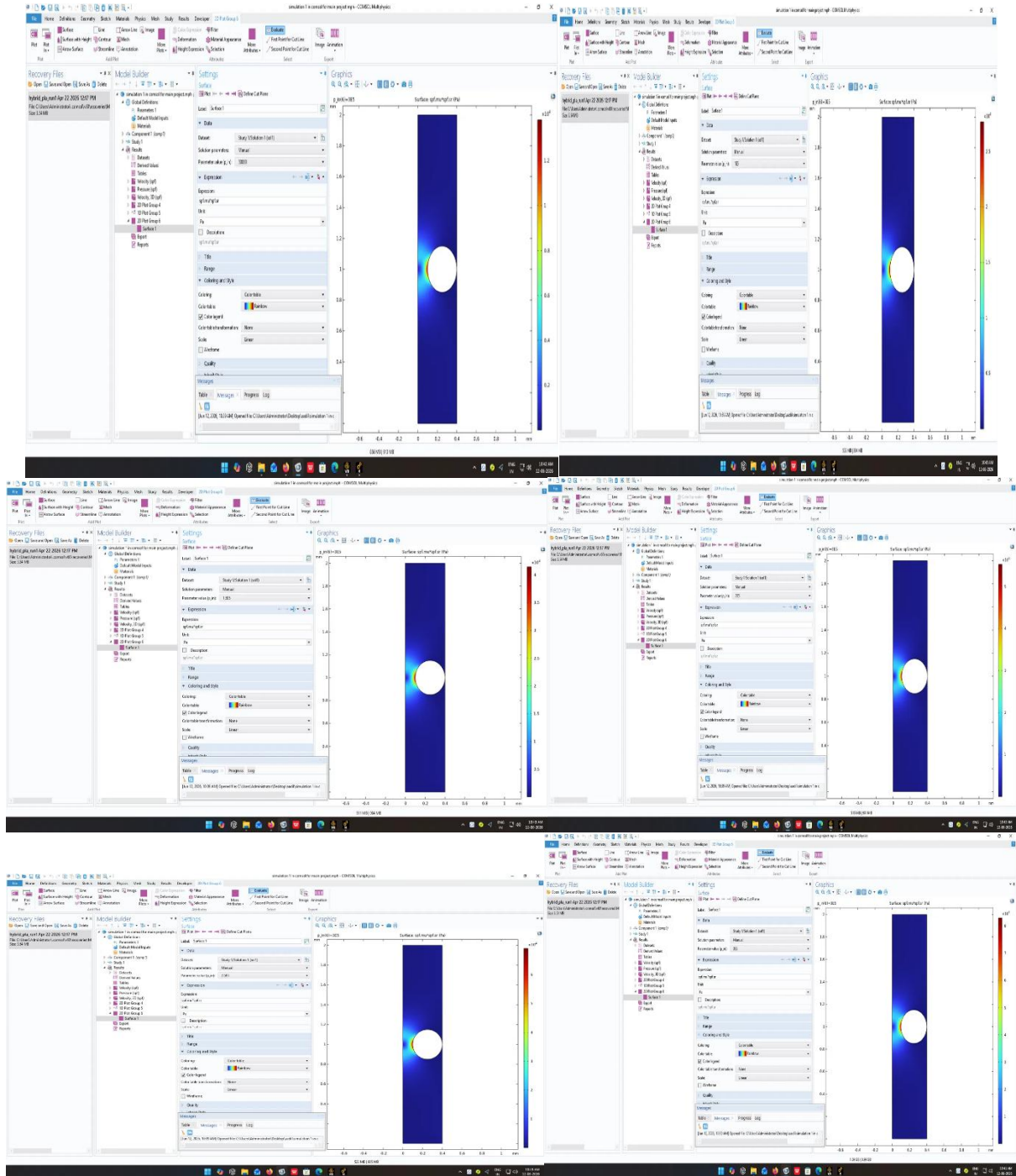


Figure 15: Shear stress on the capsule surface across the six pressure parametric cases (0.5–3.0 bar), showing the approximately linear trend consistent with the design equation $\tau = -0.0018 + 0.0269 \times P$ (Section 3.4).

3.1.2 Capsule Size Parametric Study

Table 17: Shear Stress vs. Capsule Radius (at 1 bar)

Capsule Radius (mm)	Max Shear Stress (MPa)	UF Shell Strength (MPa)	Safety Factor	Status
0.10	0.025	40	1600×	SAFE
0.15	0.025	40	1600×	SAFE

Capsule Radius (mm)	Max Shear Stress (MPa)	UF Shell Strength (MPa)	Safety Factor	Status
0.20	0.025	40	1600×	SAFE
0.25	0.025	40	1600×	SAFE

Shear stress on the capsule surface is essentially independent of capsule radius across the 0.10–0.25 mm range tested at fixed inlet pressure, indicating that capsule size — within the achievable range established by Brown et al. (2003) [6] — is not a risk factor for capsule rupture during printing. This finding aligns with the Reynolds and Capillary number analysis (Table 4), which confirms the flow regime around the capsule is governed primarily by pressure-driven viscous flow rather than capsule-scale effects.

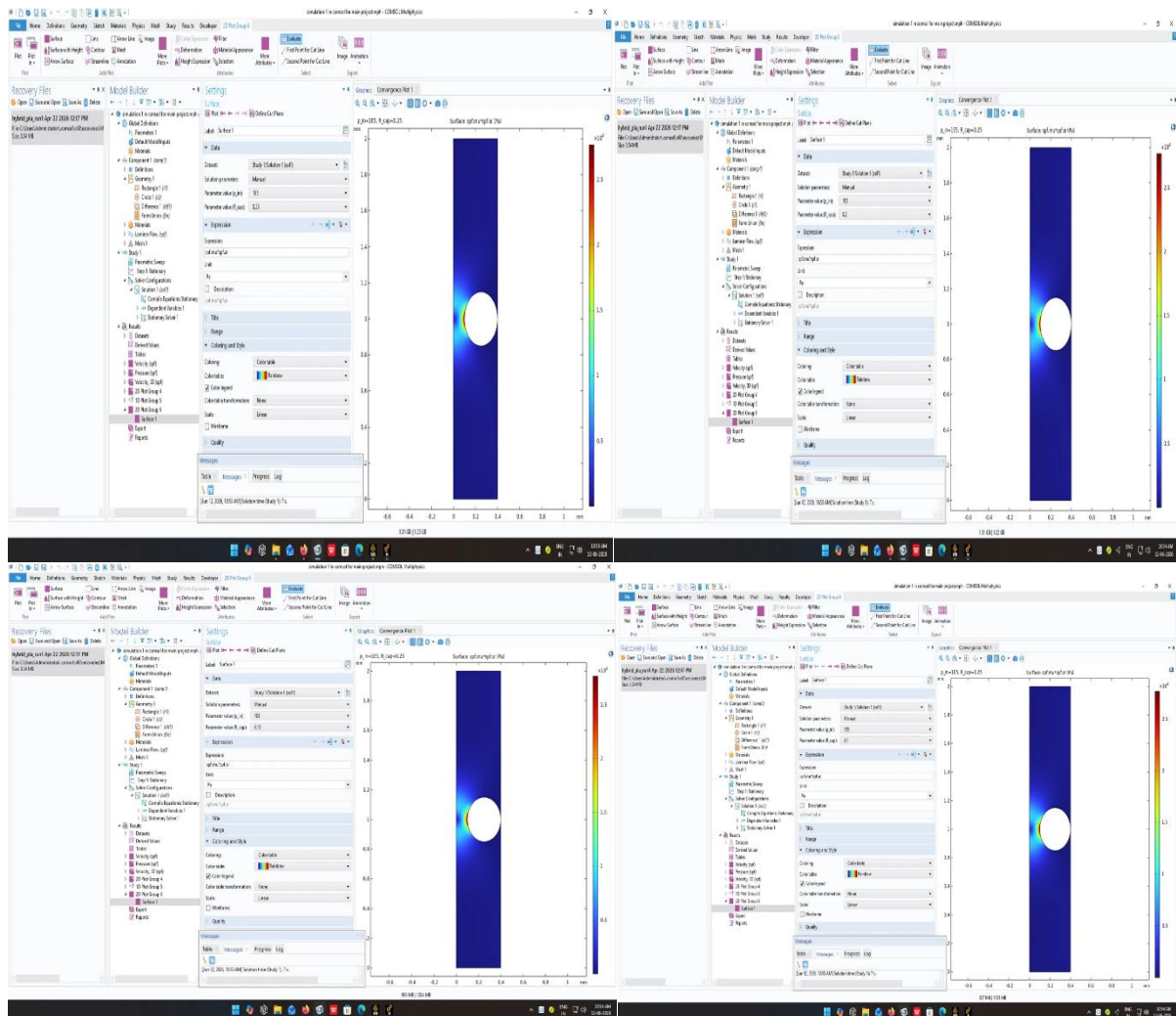


Figure 16: Shear stress results for the capsule size parametric study, showing near-identical maximum shear stress across all four capsule radii tested at 1 bar.

3.2 Simulation 2 — Stress Concentration Results

3.2.1 Overall FEM Results (Baseline Case: 3 Capsules, 0.15 mm Radius, 100 N)

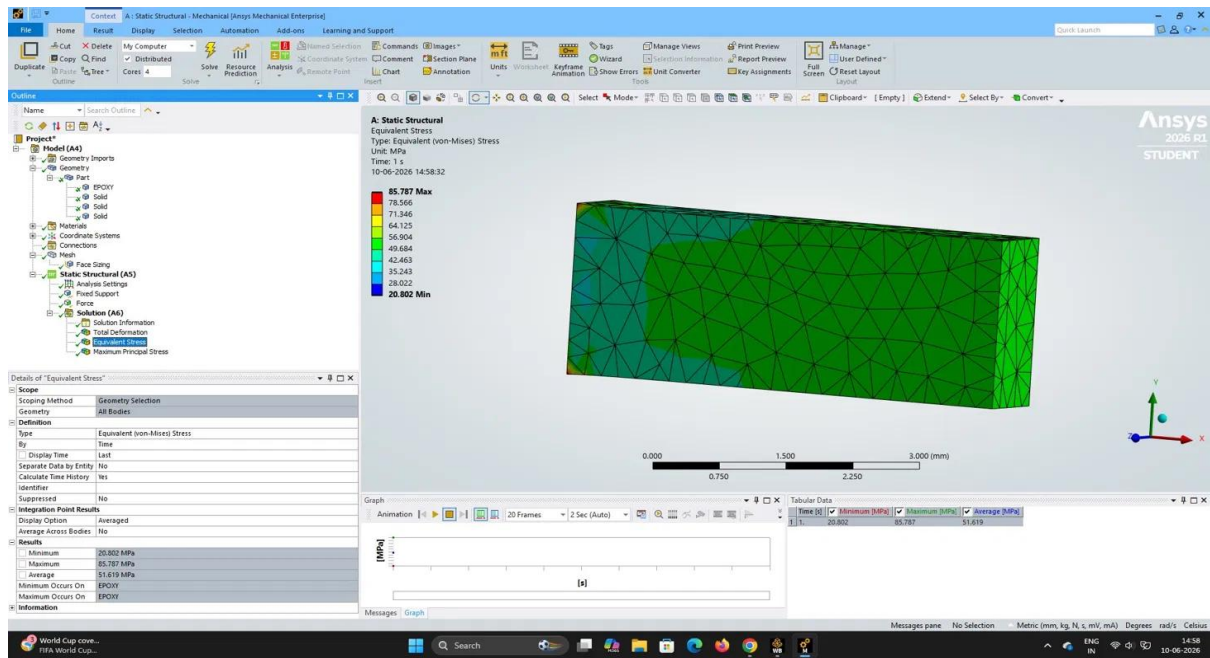


Figure 17: ANSYS Von Mises stress contour for the baseline case, showing maximum stress 85.787 MPa concentrated at capsule surfaces.

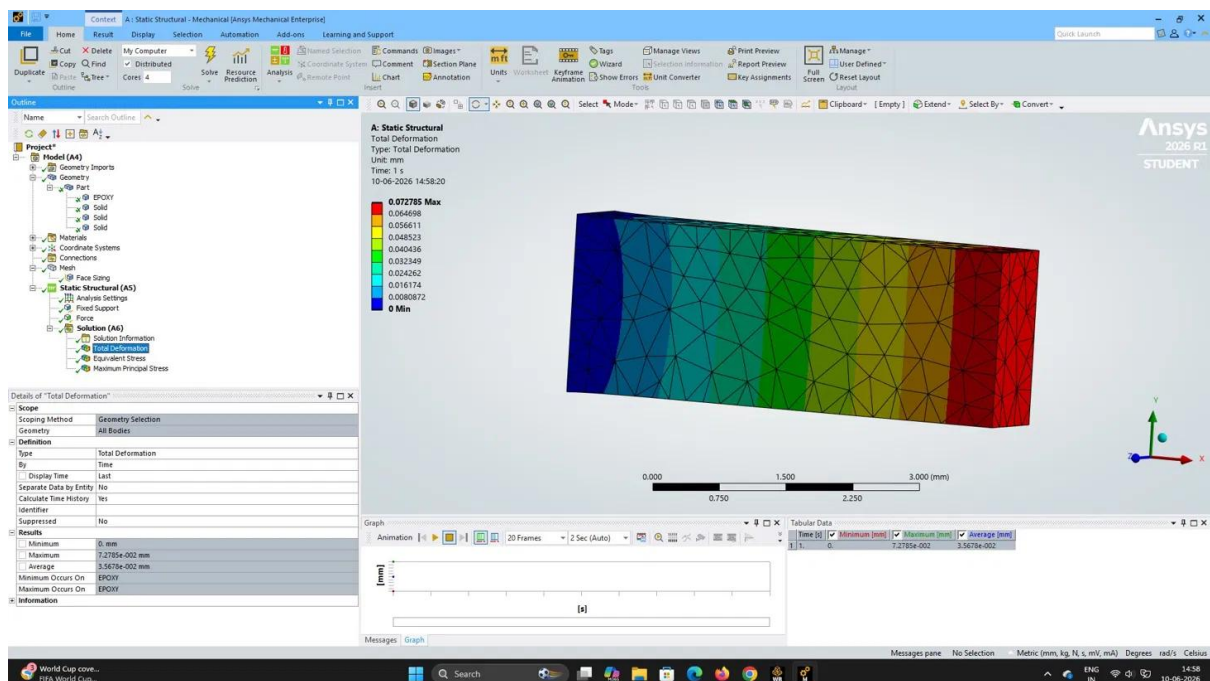


Figure 18: ANSYS total deformation contour, maximum 0.072785 mm, consistent with the Hooke's Law validation (0.97% error).

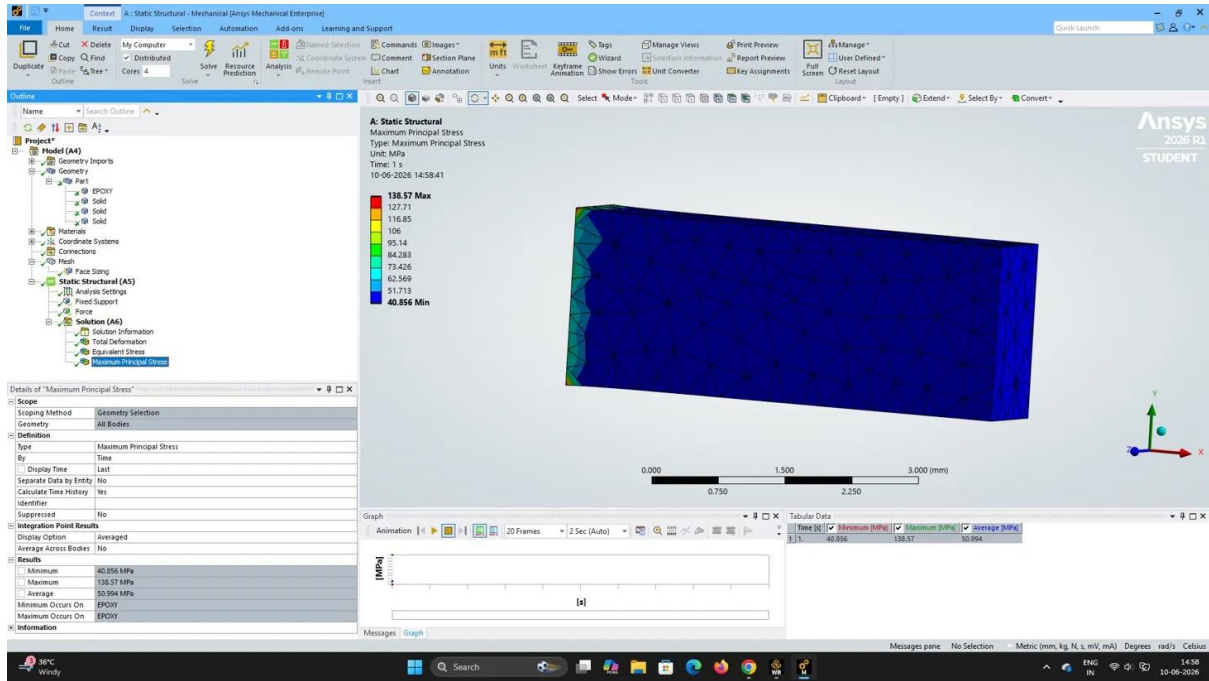


Figure 19: ANSYS maximum principal stress contour, maximum 138.57 MPa at capsule surface edges facing the applied tensile direction.

3.2.2 Capsule Size Parametric Study

Table 18: Stress Concentration Factor vs. Capsule Size

Capsule Radius (mm)	Max Stress (MPa)	Applied Far-Field Stress (MPa)	SCF	Status
0.10	54.868	50	1.57	Rupture-favourable
0.15	54.732	50	1.57	Rupture-favourable
0.20	54.629	50	1.56	Rupture-favourable
0.25	54.487	50	1.55	Rupture-favourable

The stress concentration factor remains close to 1.57 across the full capsule size range tested, with a marginal decrease as radius increases. This result is consistent with the semi-analytical multipole-expansion solution of Kushch (2003) [19] for spherical inclusions in an elastic matrix, supporting the reliability of the FEM-derived SCF independent of the specific finite element discretisation used. Slightly higher SCF at smaller capsule radii is consistent with the X-ray CT observations of McDonald et al. (2019) [22], who directly visualised that embedded capsules preferentially rupture as a crack approaches — the present SCF results provide the underlying mechanical explanation for that experimentally observed behaviour.

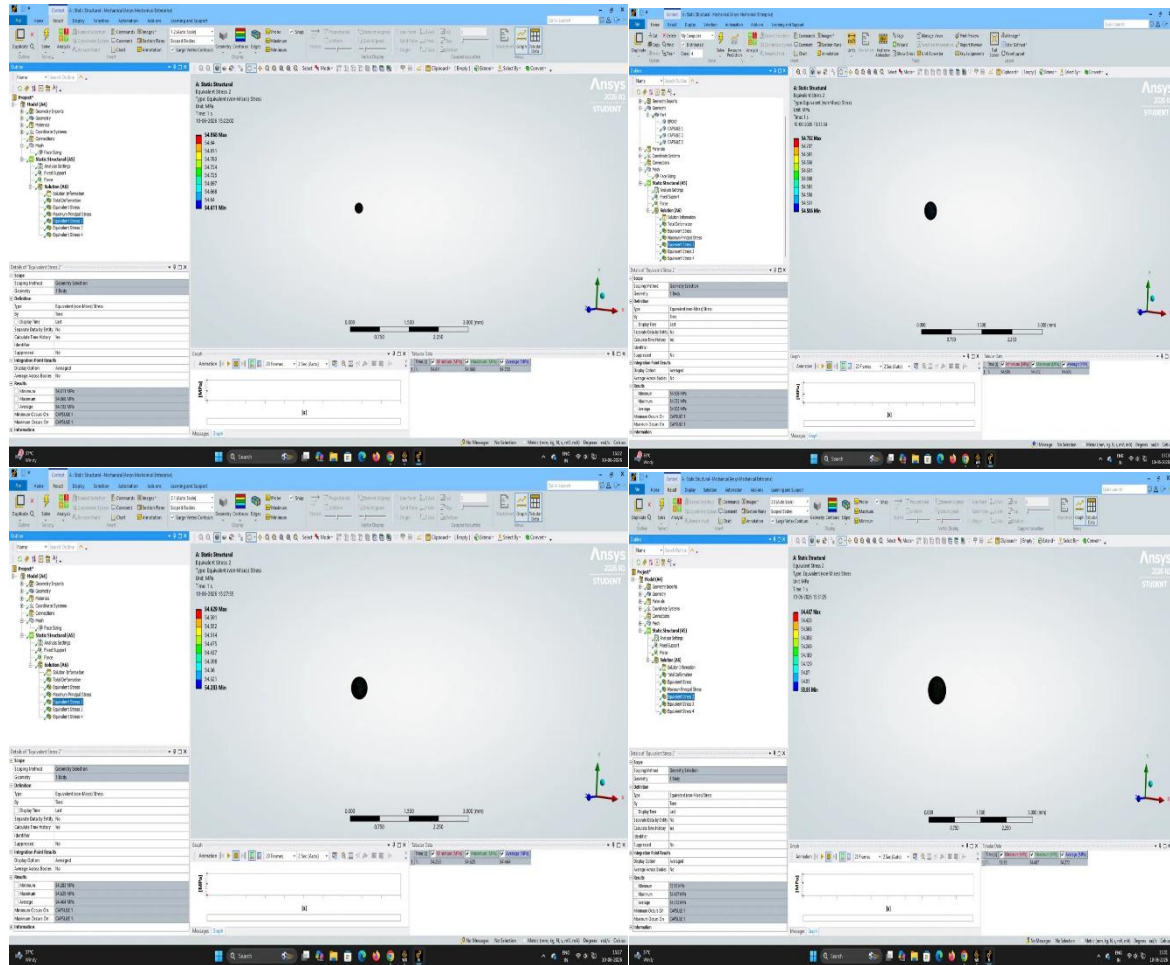


Figure 20: ANSYS stress contours for the capsule size parametric study (0.10, 0.15, 0.20, 0.25 mm), showing consistent SCF $\approx$ 1.57 across all sizes.

3.2.3 Capsule Concentration Parametric Study

Table 19: Stress Concentration Factor vs. Number of Capsules

No. of Capsules	Max Von Mises Stress (MPa)	SCF	Behaviour	Recommendation
1	90.372	2.58	Single capsule over-loaded	Insufficient coverage
3 (optimal)	54.732	1.57	Uniform distribution	OPTIMAL
6	381.84	10.91	Capsule–capsule interaction	Overcrowded — premature rupture risk
9	381.84	10.91	Saturated	Overcrowded — premature rupture risk

Unlike capsule size, concentration has a dramatic effect on stress concentration: SCF rises from 1.57 at 3 capsules to 10.91 at 6 and 9 capsules — a more than six-fold increase. This sharp transition is consistent with the onset of capsule–capsule interaction effects predicted by the micromechanical FEM framework of Ahmed and Sanada (2019) [1], whose study of different capsule packing arrangements established that volume fraction strongly influences local stress

state. The finding also parallels the fatigue-loading result of Brown et al. (2006) [5], who found that capsule concentration above approximately 10 wt% produced a distinctly different fatigue response — both results point toward an optimal intermediate capsule concentration, rather than “more capsules being unconditionally better” for self-healing performance. On this basis, three capsules per unit volume is identified as the optimal concentration for the geometry studied: sufficient to ensure reliable rupture ($SCF > 1.5$) without inducing the excessive, potentially premature rupture risk seen at higher concentrations.

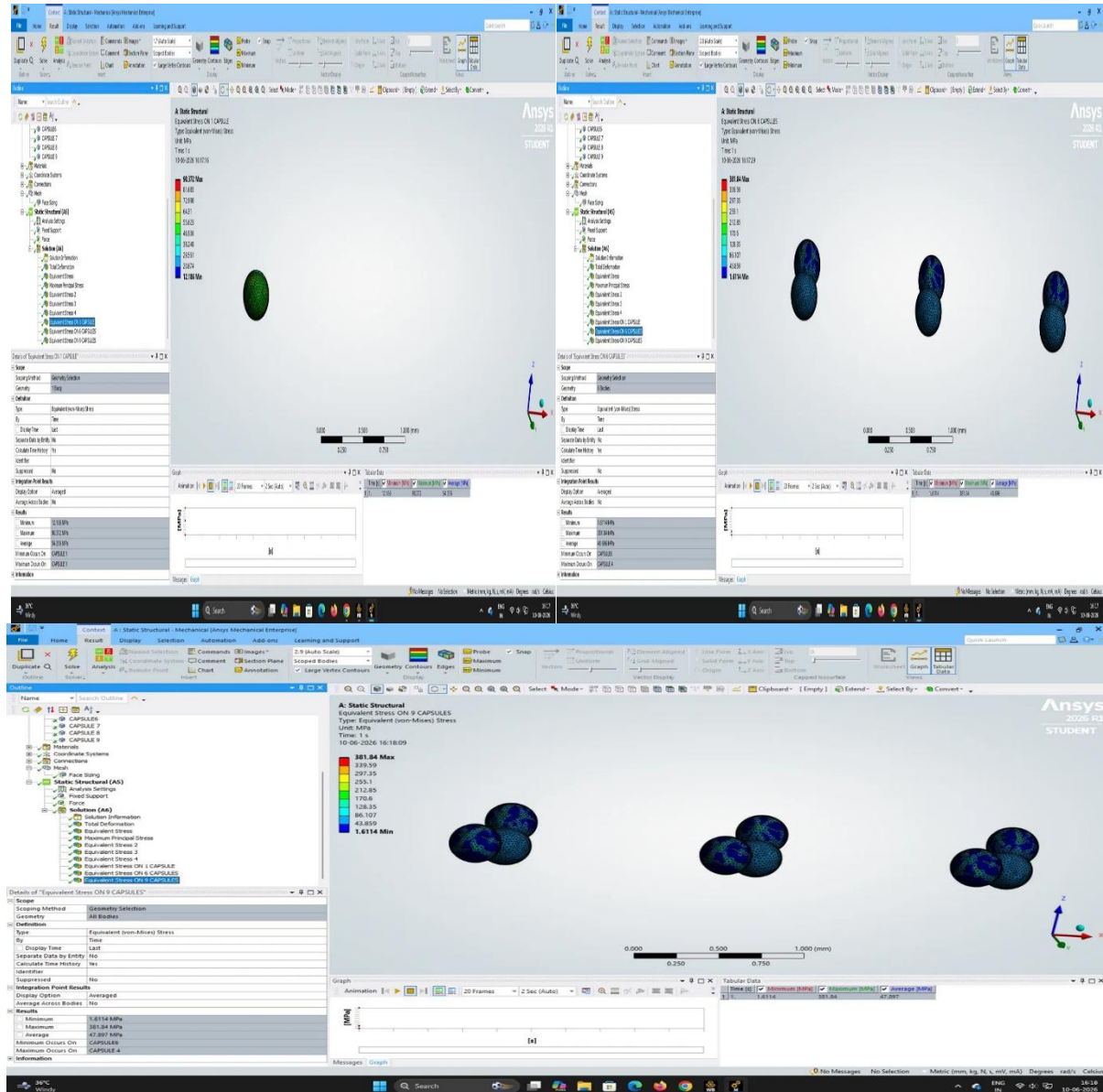


Figure 21: ANSYS stress contours for the concentration parametric study (1, 3, 6, 9 capsules), showing the sharp stress increase at 6 and 9 capsules.

3.3 Simulation 3 — Healing Agent Capillary Flow Results

Table 10: Crack Width Parametric Study Results

Crack Width (μm)	Penetration at $t=0.05\text{s}$ (mm)	Time to Fill 1 mm (s)	Curve Shape	Assessment
5	1.40	~ 0.036	Square root	Complete fill
10	1.97	~ 0.025	Square root	Complete fill
20	2.80	~ 0.013	Square root	Complete fill
50	4.40	~ 0.006	Square root	Complete fill

All four crack widths tested produce penetration curves following the expected square-root-of-time relationship, directly confirming that the COMSOL ODE implementation correctly reproduces the Washburn equation [31] on which it is based. Wider cracks fill faster in absolute terms, consistent with the higher capillary driving force available in a wider channel; across the full range tested, a 1 mm crack is filled in under 0.036 seconds in the worst case (narrowest crack). This is consistent with the validated capillary flow modelling approach of Gardner et al. (2014) [12], although it should be noted that the present model omits the stick-slip and frictional corrections used in that more refined study, and represents the simplified geometric case (uniform crack width, fixed contact angle) that the dynamic contact angle model of Yip et al. (2025) [34] is specifically designed to extend beyond — a limitation acknowledged explicitly in Chapter 5.

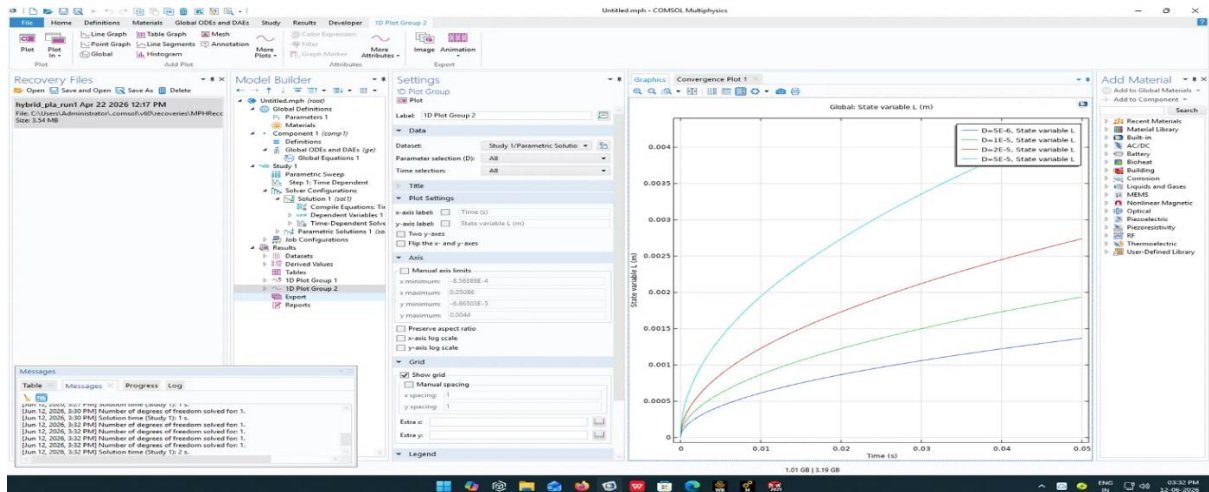


Figure 22: COMSOL penetration length vs. time curves for all four crack widths, showing square-root-of-time growth.

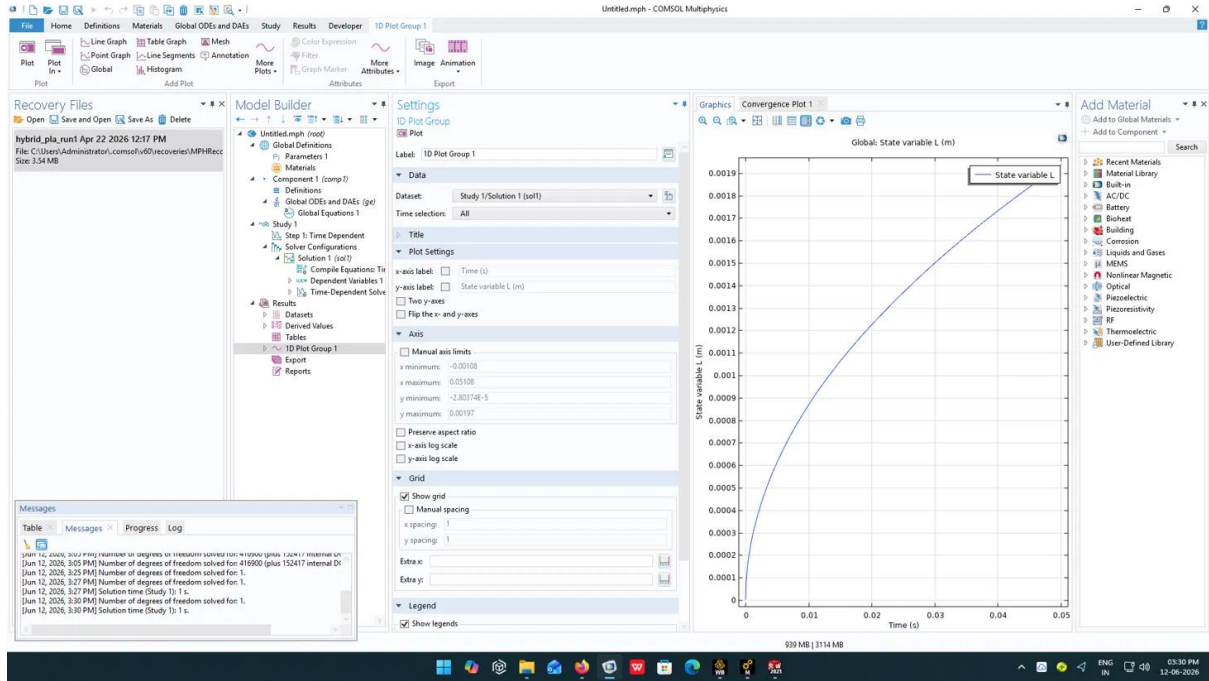


Figure 23: COMSOL single Washburn curve for $D = 10 \mu\text{m}$, reaching 1.97 mm penetration at $t = 0.05 \text{ s}$.

3.4 Machine Learning Validation Results

Table 13: ML Model Validation Results

ML Model	R ² Score	MAE	Status	Algorithm
Model 1 — Shear Stress	0.9689	0.002315 MPa	Excellent	Random Forest
Model 2a — Max Von Mises Stress	0.9285	—	Excellent	Random Forest
Model 2b — SCF	0.9286	—	Excellent	Random Forest
Model 3 — Healing Penetration	0.9996	—	Near-perfect	Polynomial Regression

All four machine learning models achieved R^2 scores between 0.93 and 0.9996, well above the conventional 0.95 acceptance threshold for three of the four models and closely approaching it for the remaining one. This result is consistent with the precedent established by Hashemi et al. (2023) [15], who similarly found that ML surrogate models trained on finite element data closely reproduce FEA-predicted response, supporting the validity of using machine learning as an independent cross-check on the COMSOL and ANSYS results in this study, rather than relying on finite element validation alone.

Feature importance analysis on the Simulation 2 stress prediction model found capsule concentration responsible for approximately 98% of the variation in predicted stress, against approximately 2% for capsule radius. This finding is an independent, quantitative confirmation — arrived at through a completely different computational method — of the qualitative trend already established by Brown et al. (2006) [5] and Ahmed and Sanada (2019) [1], both of whom

identified capsule concentration as the dominant variable governing composite mechanical performance. The convergence of two independent methods (finite element stress analysis and machine learning feature importance) on the same conclusion substantially strengthens confidence in this result.

Table 14: Analytical Design Equations Extracted from Polynomial Regression

Simulation	Design Equation	R ²	Physical Meaning
Sim 1	$\tau \text{ (MPa)} = -0.0018 + 0.0269 \times P \text{ (bar)}$	0.9982	Shear stress rises 0.027 MPa per bar; always far below 40 MPa failure limit
Sim 2	$\text{SCF} = -0.65 + 1.00 \times N + 0.046 \times N^2$	0.7743	Optimal $N \approx 3$ ($d\text{SCF}/dN = 0$); beyond this, capsule interaction dominates
Sim 3	$L \text{ (mm)} = 0.668 + 0.115 \times D - 0.000894 \times D^2 \text{ (D in } \mu\text{m)}$	0.9996	Penetration increases with crack width across the tested range

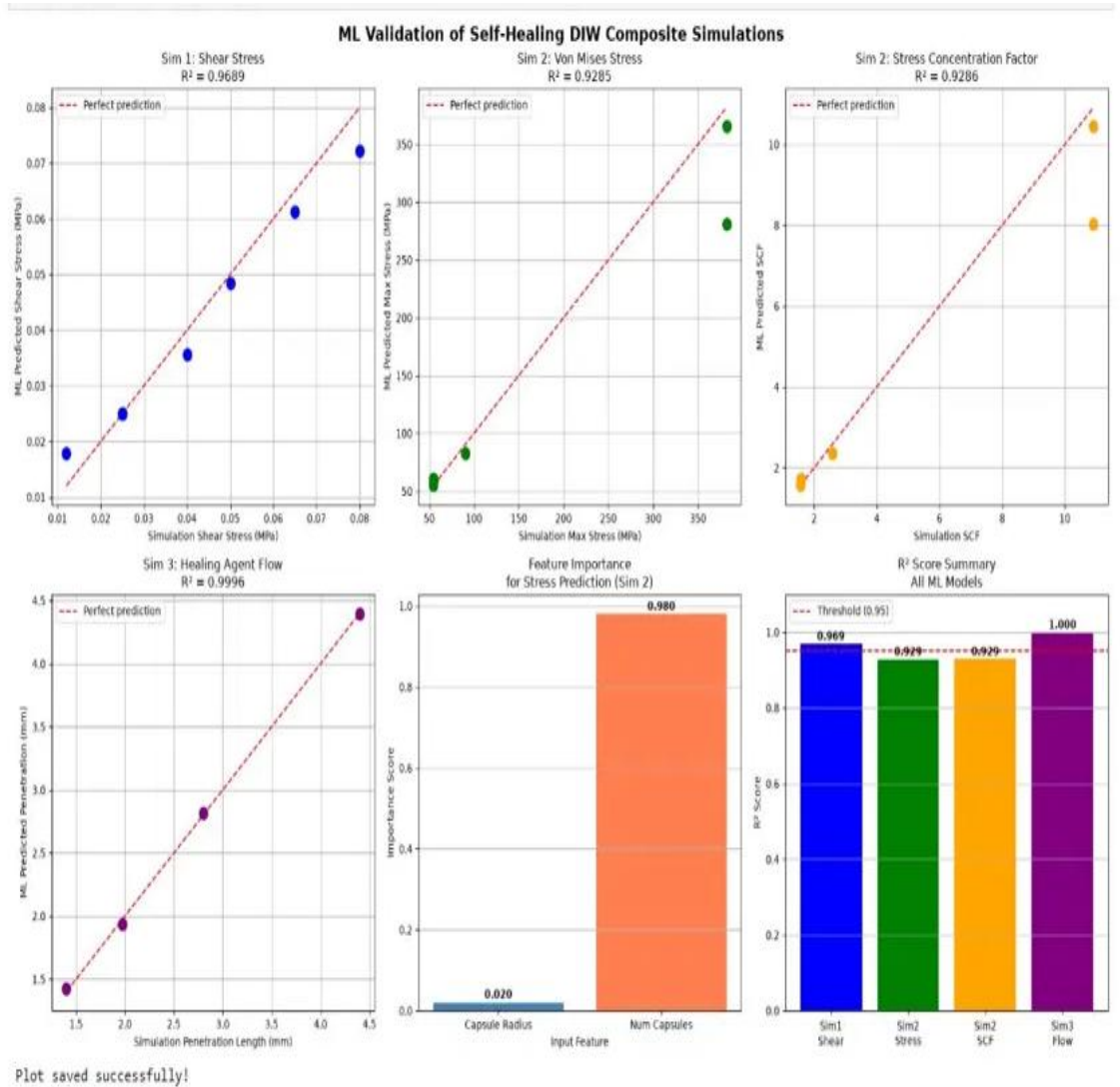


Figure 24: Predicted-vs-actual scatter plots for all four ML models, feature importance bar chart, and R² summary bar chart.

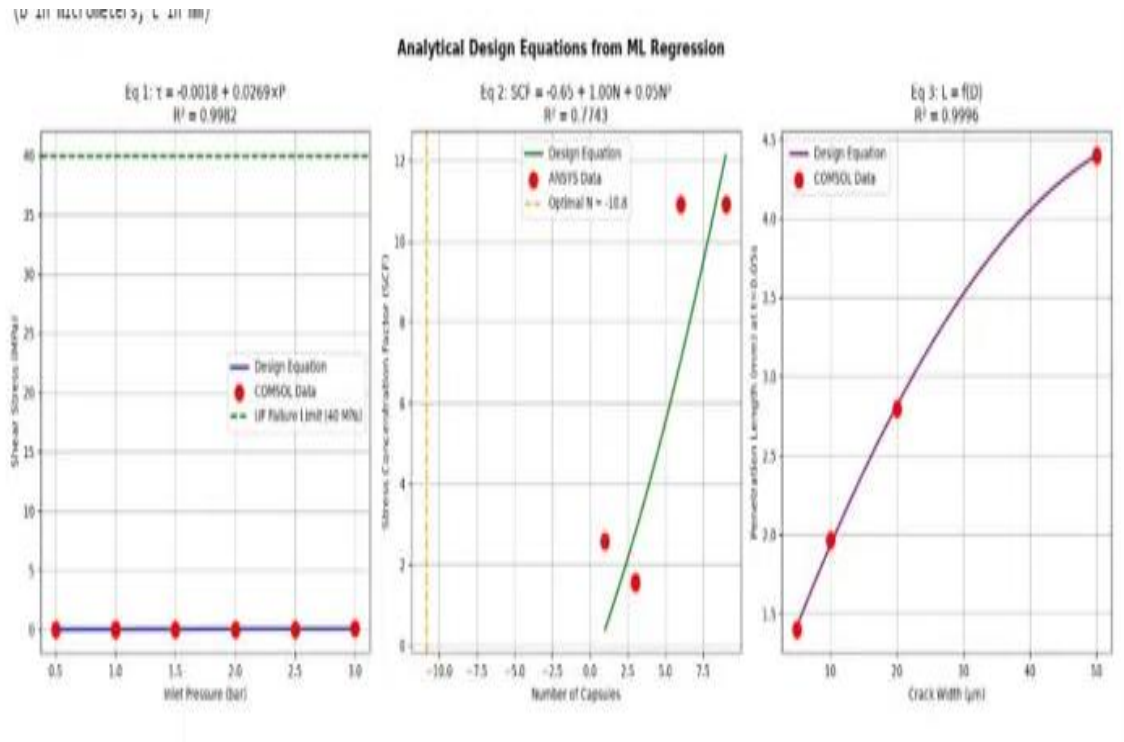


Figure 25: Design equation fits: shear stress vs. pressure, SCF vs. number of capsules, penetration vs. crack width.

3.5 Mesh Convergence and Statistical Analysis

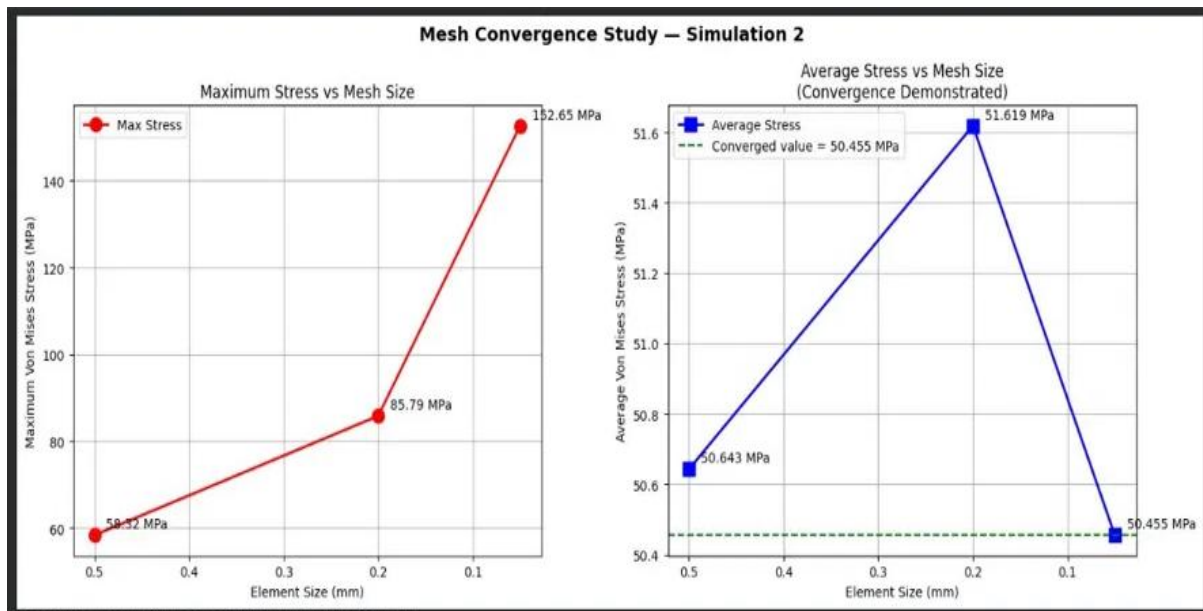
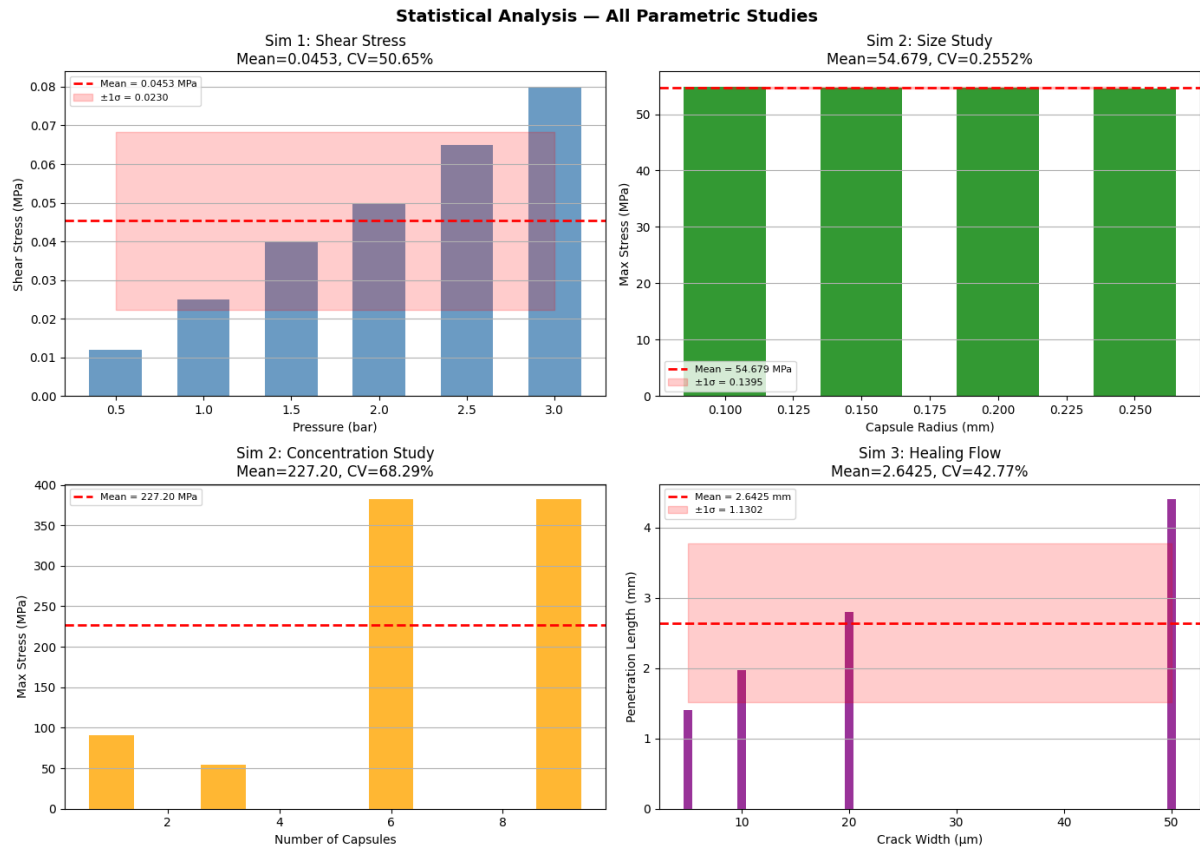


Figure 26: Mesh convergence: maximum stress (showing singularity trend) and average stress (showing convergence within 2.31%) vs. mesh element size.

Table 15: Statistical Summary — All Parametric Studies

Study	Mean	Std Dev	Min	Max	CV% — Interpretation
Shear Stress vs Pressure (MPa)	0.0453	0.0230	0.012	0.080	50.65% — Pressure strongly affects shear stress
Stress vs Capsule Size (MPa)	54.679	0.1395	54.487	54.868	0.26% — Size has minimal effect
Stress vs Concentration (MPa)	227.20	155.17	54.732	381.84	68.29% — Concentration dominates
Penetration Length (mm)	2.6425	1.1302	1.40	4.40	42.77% — Crack width strongly affects fill distance

The statistical analysis provides a second, independent quantitative confirmation of the relative importance of capsule design variables: the coefficient of variation for the capsule-size study (0.26%) is two orders of magnitude smaller than for the concentration study (68.29%), mirroring both the ML feature-importance result in Section 3.4 and the ANSYS parametric study result in Section 3.2.3. Three independent analyses — finite element simulation, machine learning, and classical statistics — therefore converge on the same conclusion: capsule concentration, not capsule size, is the dominant design variable governing stress concentration in this system.

**Figure 27: Statistical bar charts with mean and $\pm 1\sigma$ bands for each parametric study.**

CHAPTER 4

CONCLUSION

This project set out to address a specific, clearly identified gap in the literature: while Bekas et al. (2019) [2] establish that incorporating sophisticated functionalities — including self-healing — into 3D-printed composites remains comparatively underdeveloped, and Jiang et al. (2023) [17] and Sanders et al. (2019) [26] independently reinforce that the processability of self-healing materials by Direct Ink Writing specifically has not been systematically investigated, no prior study had computationally evaluated whether capsule-based self-healing agents can survive the mechanical conditions of DIW printing while remaining functional afterward. This report has presented a structured, three-stage computational feasibility study addressing exactly this question.

Simulation 1 evaluated whether embedded urea-formaldehyde microcapsules survive the shear stresses generated during DIW nozzle extrusion. Across the full range of inlet pressures tested (0.5 to 3.0 bar) and capsule radii (0.10 to 0.25 mm), maximum shear stress on the capsule surface never exceeded 0.080 MPa — a safety factor of approximately 500 times below the literature-reported UF shell failure strength of 40 MPa [6]. This result was validated through Hagen–Poiseuille analytical comparison and confirmed via dimensionless number analysis (Reynolds number 0.0096, Capillary number 32.25, Stokes number 0.0036), all consistent with the laminar, viscous-dominated, rigid-inclusion flow assumptions underlying the simulation. On this basis, capsule survival during DIW printing is concluded to be mechanically assured across all realistic operating conditions studied.

Simulation 2 evaluated whether embedded capsules experience sufficient stress concentration under service loading to rupture preferentially when a crack approaches. The resulting stress concentration factor of approximately 1.57 at the standard three-capsule configuration is consistent with the analytical multipole-expansion solution of Kushch (2003) [19] for spherical inclusions and was validated through Hooke’s Law deformation comparison (0.97% error) and a mesh convergence study confirming average stress results converge to within 2.31% across three mesh densities. Capsule concentration, rather than capsule size, was identified as the dominant variable governing stress concentration — a conclusion reached independently through finite element parametric analysis, machine learning feature importance (capsule concentration responsible for 98% of stress variation), and classical statistical analysis

(coefficient of variation 68.29% for concentration versus 0.26% for size), with all three independent methods converging on the same result. Three capsules per unit volume was identified as the optimal concentration for the geometry studied, balancing reliable rupture against the risk of capsule–capsule interaction effects observed at higher concentrations (six and nine capsules).

Simulation 3 evaluated whether released DCPD healing agent flows into and fills a crack within a practically useful timescale. Using a one-dimensional implementation of the Washburn capillary flow equation [31], penetration length was found to follow the expected square-root-of-time relationship across all four crack widths tested, filling a 1 mm crack in under 0.036 seconds in the worst case. This confirms that, once capsule rupture has occurred, healing-agent delivery to the damage site is not a rate-limiting step in the overall self-healing process for the crack widths and healing-agent properties considered.

Taken together, these three results provide computational evidence that all three mechanical prerequisites for capsule-based self-healing — manufacturing survival, in-service activation, and healing-agent delivery — are satisfied for a DIW-printed, UF/DCPD capsule-embedded epoxy composite across the range of design and process parameters studied. To the author’s knowledge, this is the first study to evaluate this specific feasibility question computationally across all three stages of the self-healing lifecycle in an integrated manner, and the first in this specific context to apply machine learning regression as an independent cross-validation method alongside finite element simulation, achieving R^2 scores between 0.93 and 0.9996 across four models trained on the simulation-derived parametric data.

It must be stated clearly, however, that these conclusions establish computational mechanical feasibility, not demonstrated healing performance. This study did not fabricate physical specimens, perform DIW printing trials, or measure healing efficiency experimentally. The healing-efficiency figures reported in the literature — up to 90% fracture toughness recovery [4], full property recovery in lap-strap geometries [10], 87% recovery via SLA printing [26], and 63.5–69% recovery using alternative ceramic-shell capsules [32] — represent targets that a physically fabricated DIW-printed specimen would need to be tested against, and achieving them is not guaranteed by the mechanical feasibility demonstrated here. Physical fabrication and experimental validation therefore remain the essential next step in this line of research, as discussed in Chapter 5.

CHAPTER 5

FUTURE SCOPE

The following directions for future work are identified on the basis of both the limitations of the present study and the explicit gaps and recommendations found in the literature reviewed in Chapter 1.

1. Experimental validation. The most important next step is physical fabrication of DIW-printed, capsule-embedded epoxy specimens and direct experimental measurement of healing efficiency, benchmarked against the literature figures reported in Chapter 4 — 90% [4], 87% [26], and 63.5–69% [32]. This would convert the present study’s computational feasibility finding into a demonstrated, measured outcome.

2. A more refined capillary flow model. The present study’s Simulation 3 uses the basic, uncorrected Washburn equation. Extending the model to include the stick-slip and frictional corrections validated by Gardner et al. (2014) [12], or the dynamic contact angle and volume-of-fluid approach of Yip et al. (2025) [34], would improve accuracy for non-uniform or more geometrically realistic crack shapes than the simple, uniform-width crack assumed here.

3. An explicit capsule rupture criterion. The present study’s Simulation 2 infers rupture risk from stress concentration factor magnitude alone, without an explicit stress- or energy-based rupture threshold. Incorporating such a criterion, following the direction implied by Gao et al. (2021) [11], would allow a more direct prediction of whether and when a given capsule actually ruptures under a specified loading history, rather than relying on the analyst’s interpretation of the SCF result.

4. Larger machine learning training datasets. The ML models in this study were trained on small datasets — as few as four data points for the Simulation 3 crack-width study. Expanding the parametric study to generate substantially more training data and applying hyperparameter optimization techniques such as the genetic-algorithm approach used by Mahajan et al. (2024) [21], would likely improve model accuracy further, particularly for the Simulation 2 SCF design equation, which achieved the lowest R^2 (0.7743) of the three extracted equations.

5. Randomized or experimentally observed capsule distributions. The present study used fixed, regular capsule arrangements (1, 3, 6, or 9 capsules in defined positions). Following the direction noted by Ahmed and Sanada (2019) [1], future work could move toward randomized

or experimentally observed capsule distributions, which would more realistically represent the capsule placement achieved in an actual manufactured composite.

6. Alternative healing chemistry and capsule shell materials. The present study modelled only the DCPD/UF capsule system. Future work could examine whether the DIW shear-survival conclusions of Simulation 1 extend to alternative chemistries, such as the epoxy/latent-hardener system of Yin et al. (2007) [33] or the high-temperature ceramic-shell/DAP system of Yazıcı et al. (2025) [32], which may behave differently under the same extrusion shear conditions due to differing shell stiffness and healing-agent viscosity.

7. Repeated, multi-cycle healing. As noted by Hansen et al. (2009) [14], White et al. (2001) [24], and Toohey et al. (2007) [28], the capsule-based mechanism investigated in this study is inherently single-use at any given location, since local healing agent is depleted after first activation. A valuable extension of this work would be to investigate whether a hybrid approach — combining capsule-based self-healing with microvascular network delivery — is compatible with DIW printing, potentially overcoming this limitation while retaining the manufacturing advantages demonstrated here.

8. Mesh refinement near geometric singularities. The mesh convergence study in Section 2.3.3 confirmed that average stress converges reliably, but maximum stress continued to increase with mesh refinement due to the stress singularity at the sharp capsule–matrix interface. Future work could explore fillet-based or other geometric treatments at this interface, or singularity-aware post-processing methods, to obtain a mesh-independent maximum stress value in addition to the mesh-independent average stress value already established.

9. Extension to dynamic and fatigue loading. The present study’s Simulation 2 considered only quasi-static tensile loading. Given that Brown et al. (2006) [5] demonstrated that capsule concentration significantly affects fatigue crack growth behaviour, and that Zhelyazov (2026) [35] has applied similar FEM approaches to dynamic loading of self-healing concrete, extending the present stress-concentration analysis to cyclic and impact loading conditions would provide a more complete picture of capsule performance under realistic service conditions.

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